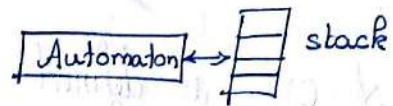
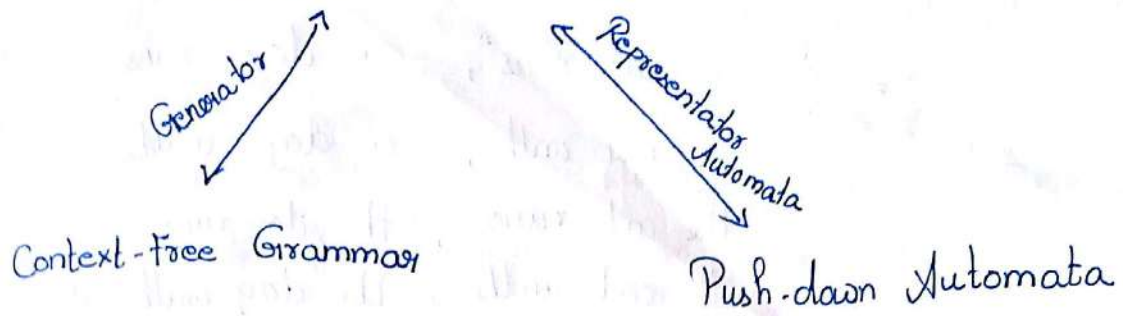
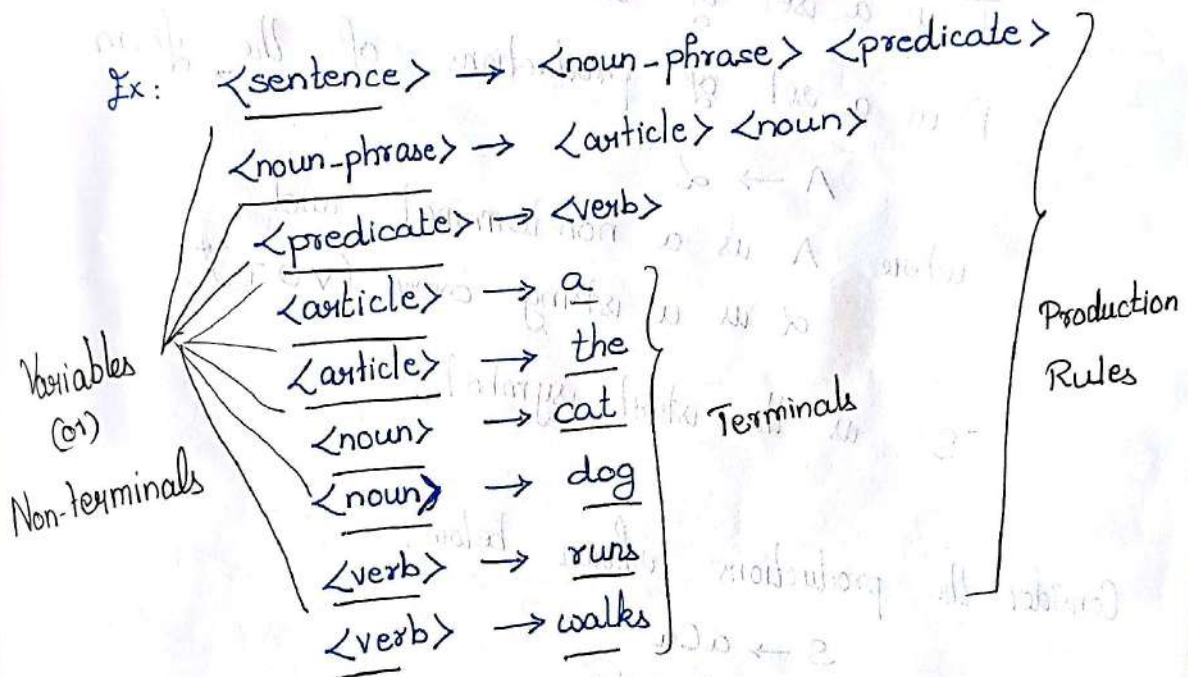


Context-Free Languages



Grammars:

Grammars express languages.



Derivation of "The dog walks"

$\langle \text{sentence} \rangle \rightarrow \langle \text{noun-phrase} \rangle \langle \text{predicate} \rangle$
 $\rightarrow \langle \text{noun-phrase} \rangle \langle \text{verb} \rangle$
 $\rightarrow \langle \text{article} \rangle \langle \text{noun} \rangle \langle \text{verb} \rangle$
 $\rightarrow \underline{the} \langle \text{noun} \rangle \langle \text{verb} \rangle$
 $\rightarrow \underline{the} \underline{dog} \langle \text{verb} \rangle$
 $\rightarrow \underline{the} \underline{dog} \underline{walks}$

Language of the Grammar:

$L = \{ \text{"a cat runs", a dog runs, a cat walks, a dog walks, the cat runs, the dog runs, the cat walks, the dog walks} \}$

A CFG is defined as $G = (V, T, P, S)$ where

V is a set of non-terminals

T is a set of terminals

P is a set of productions of the form

$$A \rightarrow \alpha$$

where A is a non-terminal and

α is a string over $(V \cup T)^*$

S is the start symbol.

Consider the productions shown below:

$$S \rightarrow aCa$$

$$C \rightarrow aCa/b$$

Grammar $G = (V, T, P, S)$

$$V = \{S, C\}$$

$$T = \{a, b\}$$

$$P = \{ S \rightarrow aCa, C \rightarrow aCa/b \}$$

S is the start symbol.

④ Let $G = (V, T, P, s)$ be a Grammar. Then the set $L(G) = \{w \in T^* : s \xRightarrow{*} w\}$ is the language generated by G .

⑤ The strings which contain variables as well as terminals are called sentential forms of the derivation.
(or) sequential forms.

⑥ Derivation Tree. (or) Parse Tree ⑦ Yield.

① Consider the grammar $G = (\{S\}, \{a, b\}, S, P)$ and P is given by

$$S \rightarrow aSb \mid \epsilon$$

$$\begin{aligned} \omega_1: S &\Rightarrow aSb & \omega_2 \\ &\Rightarrow aaSbb & \omega_3 \\ &\Rightarrow aabb & \omega_4 \end{aligned}$$

$$\boxed{S \xRightarrow{*} aabb}$$

$$\omega_1 \xRightarrow{*} \omega_4.$$

$\xRightarrow{*}$ derivation
 $\xRightarrow{d}$ leftmost derivation
 $\xRightarrow{r}$ rightmost derivation

$$\omega_1 \Rightarrow \omega_2 \Rightarrow \omega_3 \Rightarrow \dots \Rightarrow \omega_n$$

$$\omega_1 \xRightarrow{*} \omega_n.$$

$aabb$ is the string in the CFL generated by CFG.
 $aaSbb$ is a sentential form.

Derivations:

$$S \xRightarrow{*} \epsilon \quad S \xRightarrow{*} ab \quad S \xRightarrow{*} aabb \quad S \xRightarrow{*} aaabbb$$

$$\therefore \text{Language} = \{a^n b^n \mid n \geq 0\}.$$

② Consider the Grammar $G = (V, T, P, S)$ $V = \{S\}$

$$T = \{a, b, c\}$$

$$P = \{S \rightarrow aSa \mid bSb \mid c\}.$$

Write the language generated by the Grammar.

$$\begin{aligned} S &\xRightarrow{*} c & S &\xRightarrow{*} aca & S &\xRightarrow{*} bcb & S &\xRightarrow{*} abcba \\ S &\xRightarrow{*} bacab & S &\xRightarrow{*} aacaa & S &\xRightarrow{*} bbcb b & \end{aligned}$$

$$S \xRightarrow{*} abacaba$$

$$\therefore L = \{wcw^R \mid w \in (a, b)^*\}.$$

③. $G = (V, T, P, S)$
 $V = \{s, c\}$ $T = \{a, b\}$
 $P = \{$

$$S \rightarrow aCa$$

$$C \rightarrow aCa \mid b$$

s is the start symbol. What is the language generated by this grammar?

$$s \xRightarrow{*} aba \quad s \xRightarrow{*} aabaa \quad s \xRightarrow{*} aaabaaa$$

$$L(G) = \{a^n b a^n \mid n \geq 1\}$$

④. $L = \{aa, ab, ba, bb\}$ $CFG = ?$
 $R.E = (a+b)(a+b)$

$P \Rightarrow$
 $S \rightarrow AA$ s start symbol
 $A \rightarrow a \mid b$ $T = \{a, b\}$ $N = \{S, A\}$

⑤. $L = \{a^n \mid n \geq 0\}$
 $\{a, aa, aaa, \dots\}$

$P \Rightarrow$
 $S \rightarrow aS \mid \epsilon$ (or) $S \rightarrow Sa \mid \epsilon$
 $T = \{a\}$ $N = \{S\}$ s start symbol.

Left Recu

⑥. $(a+b)^*$

$$S \rightarrow as \mid bs \mid \epsilon$$

⑦. Atleast 2 length $(a+b)(a+b)(a+b)^*$

$$S \rightarrow AAB$$

$$A \rightarrow a \mid b$$

$$B \rightarrow aB \mid bB \mid \epsilon$$

- ⑧. Strings having length at most 2.
 $(a+b+\epsilon)(a+b+\epsilon)$.

$$S \rightarrow AA$$

$$A \rightarrow a|b|\epsilon$$

- ⑨. $a(a+b)^*b$

$$S \rightarrow aAb$$

$$A \rightarrow aA|bA|\epsilon$$

- ⑩. Strings starting and ending with diff. symbols.
 Reg. Exp. $a(a+b)^*b + b(a+b)^*a$.

$$S \rightarrow aAb|bAa$$

$$A \rightarrow aA|bA|\epsilon$$

- ii. Strings starting and ending with same symbols.
 Reg. Exp. $a(a+b)^*a + b(a+b)^*b$.

$$S \rightarrow aAa|bAb|a|b|\epsilon$$

$$A \rightarrow aA|bA|\epsilon$$

- ⑫. $a^n b^n / n \geq 1$.

$$S \rightarrow aSb|ab$$

⑬

$$ww^R \cup wa w^R \cup wb w^R \leq w \in (a+b)^*$$

$$S \rightarrow aSa|bSb|a|b|\epsilon$$

⑭

Even-length strings

$$S \rightarrow BS|\epsilon$$

$$B \rightarrow AA$$

$$A \rightarrow a|b$$

- ⑮. $a^n b^m / n, m \geq 1$.

$$S \rightarrow AB$$

$$A \rightarrow aA|a$$

$$B \rightarrow bB|b$$

$$(16). a^n b^n c^m \mid n, m \geq 1$$

$$S \rightarrow AB$$

$$A \rightarrow aAb \mid ab$$

$$B \rightarrow cB \mid c$$

$$(17). c^m a^n b^n \mid n, m \geq 1$$

$$S \rightarrow BA$$

$$A \rightarrow aAb \mid ab$$

$$B \rightarrow cB \mid c$$

$$(18). a^n c^m b^n \mid n, m \geq 1$$

$$A \rightarrow aBb \mid aAb$$

$$B \rightarrow cB \mid c$$

$$(19). a^n b^n c^m d^m \mid n, m \geq 1$$

$$S \rightarrow AB$$

$$A \rightarrow aAb \mid ab$$

$$B \rightarrow cBd \mid cd$$

$$(20). a^n b^{2n} \mid n \geq 1$$

$$S \rightarrow aSbb \mid abb$$

$$(21). a^m b^m c^m d^n \mid n, m \geq 1$$

$$S \rightarrow aSd \mid aAd$$

$$A \rightarrow bAc \mid bc$$

$$(22). a^{m+n} b^m c^n \mid m, n \geq 1$$

$$a^n a^m b^m c^n$$

$$S \rightarrow aSc \mid aAc$$

$$A \rightarrow aAb \mid ab$$

$$(23). a^n b^{n+m} c^m \mid n, m \geq 1$$

$$a^n b^n b^m c^m \mid n, m \geq 1$$

$$S \rightarrow AB$$

$$A \rightarrow aAb \mid ab$$

$$B \rightarrow bBc \mid bc$$

$$(24). a^n b^m c^{n+m} \mid n, m \geq 1$$

$$a^n b^m c^m c^n \mid n, m \geq 1$$

$$S \rightarrow aAc \mid aSc$$

$$A \rightarrow bAc \mid bc$$

$$(25). a^n b^n c^n \mid n \geq 1$$

$$(26). a^n b^m c^n d^m \mid n, m \geq 1$$

↓

CFG is not possible

$$(27). a^i b^j c^k \mid i = j + k$$

~~not possible~~

$$S \rightarrow aSc \mid T$$

$$T \rightarrow aTb \mid e$$

$$(28). \{ a^i b^j c^k \mid j = i \text{ or } j = k \}$$

$$S \rightarrow AT \mid UC$$

$$A \rightarrow aA \mid e$$

$$C \rightarrow cC \mid e$$

$$T \rightarrow bTc \mid e$$

$$U \rightarrow aUb \mid e$$

$$(29). \{a^i b^j c^k \mid i=j \text{ or } i=k\}$$

$$S \rightarrow TC \mid U$$

$$C \rightarrow cC \mid \epsilon$$

$$T \rightarrow aTb \mid \epsilon$$

$$U \rightarrow aUc \mid B$$

$$B \rightarrow bB \mid \epsilon$$

$$(30). \{a^i b^j c^k \mid i < j \text{ or } i > k\}$$

$$S \rightarrow TBC \mid AV$$

$$T \rightarrow aTb \mid \epsilon$$

$$\left\{ \begin{array}{l} B \rightarrow bB \mid b \Rightarrow \text{any no. of } b \\ C \rightarrow cC \mid \epsilon \Rightarrow \text{any no. of } c \end{array} \right\}$$

$$\left\{ \begin{array}{l} A \rightarrow aA \mid a \Rightarrow \text{any no. of } a \\ V \rightarrow aVc \mid V \\ V \rightarrow bV \mid \epsilon \end{array} \right\}$$

$$(31). \{a^i b^j \mid i \leq j \leq 2i\}$$

$$S \rightarrow aSb \mid aSbb \mid \epsilon$$

Why ^{the name} context-free Grammar?

$$A \rightarrow aAb / ab$$

$$A \Rightarrow aAb$$

$$\Rightarrow aabb$$

(irrespective of the preceding & following symbols A can be replaced with aAb or ab).

so when the productions

$$A \rightarrow \alpha \quad \begin{matrix} A \in V \\ \alpha \in (V \cup T)^* \end{matrix}$$

is called context-free Grammar.

In case...

if i have ^{a productions} $aA \rightarrow aa$

then if A has a preceding then only it will be replaced by aa .

CFG:

Can be:

Ambiguous & Unambiguous
Deterministic & Non-deterministic
Left-Recursion & Right-Recursion

①. Consider the grammar

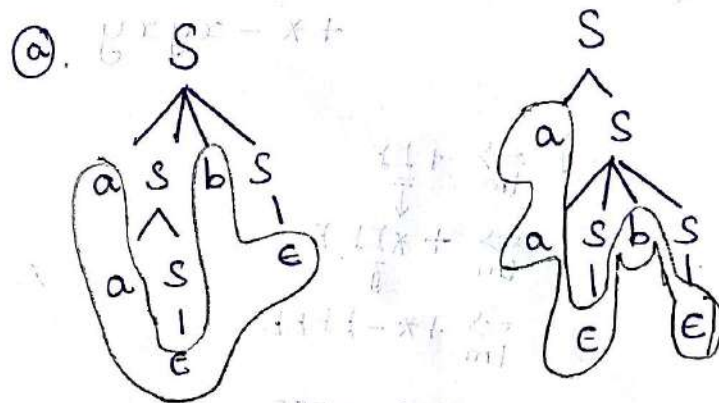
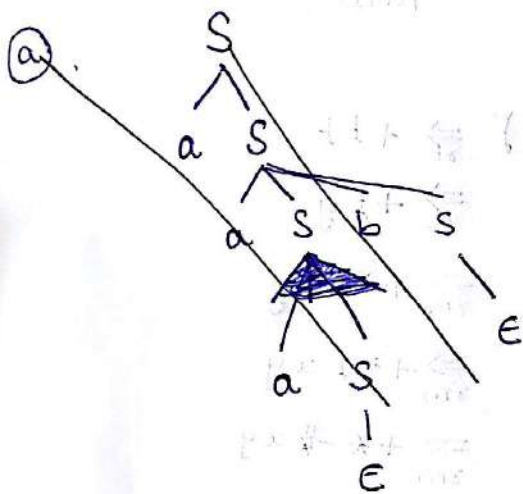
$$S \rightarrow aS \mid aSbS \mid \epsilon$$

The grammar is ambiguous. Show in particular that the string aab has two:

a). Parse trees.

b). LMD

c). RMD



③.
$$\begin{aligned} S &\Rightarrow aS \\ &\Rightarrow aSbS \\ &\Rightarrow aabS \\ &\Rightarrow aab \end{aligned}$$

$$\begin{aligned} S &\Rightarrow aSbS \\ &\Rightarrow aSbS \\ &\Rightarrow aabS \\ &\Rightarrow aab \end{aligned}$$

④.
$$\begin{aligned} S &\Rightarrow aS \\ &\Rightarrow aSbS \\ &\Rightarrow aSb \\ &\Rightarrow aab \end{aligned}$$

$$\begin{aligned} S &\Rightarrow aSbS \\ &\Rightarrow aSb \\ &\Rightarrow aSb \\ &\Rightarrow aab \end{aligned}$$

②. The following grammar generates prefix expressions with operands x and y and binary operators $+$, $-$ and $*$:

$$E \rightarrow +EE \mid *EE \mid -EE \mid x \mid y$$

②. Find LMD & RMD and derivation tree for the string

$+*-xyxy$

⑥. Prove that this grammar is unambiguous.

①. LMD

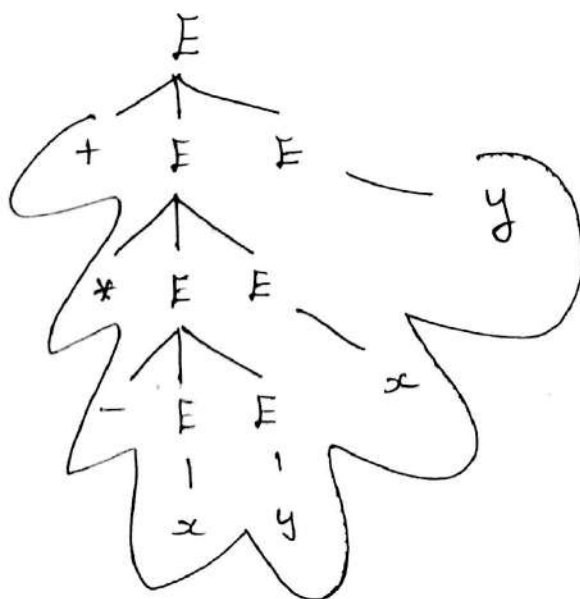
$+*-xyxy$

RMD

$$\begin{aligned} E &\Rightarrow +EE \\ &\Rightarrow +*EEE \\ &\Rightarrow +*-EEEE \\ &\Rightarrow +*-xEEE \\ &\Rightarrow +*-xyEE \\ &\Rightarrow +*-xyxE \\ &\Rightarrow +*-xyxy \end{aligned}$$

$$\begin{aligned} E &\Rightarrow +EE \\ &\Rightarrow +Ey \\ &\Rightarrow +*EEy \\ &\Rightarrow +*Exy \\ &\Rightarrow +*-Exy \\ &\Rightarrow +*-Eyx \\ &\Rightarrow +*-xyxy \end{aligned}$$

Derivation Tree



Simplified Forms and Normal Forms:

- (*)1. Elimination of Nullable Variables
- (*)2. Elimination of Unit productions
- (*)3. Elimination of Useless variables

Elimination of Nullable Variables:

Nullable Variables:

A nullable variable in a CFG $G = (V, \Sigma, S, P)$ is defined as follows...

(*) Any variable A for which P contains the production $A \rightarrow \epsilon$ is nullable.

(*) If P contains the production $A \rightarrow B_1 B_2 \dots B_n$ and $B_1, B_2, \dots, B_n$ are nullable variables, then A is nullable.

(*) No other variables in V are nullable.

Note:

If $\epsilon \in L(G)$ then $S \rightarrow \epsilon$ S cannot be removed.

Problems:

①. $S \rightarrow aSb \mid aAb$
 $A \rightarrow \epsilon$

Nullable variable: A

$$S \rightarrow aSb \mid aAb \mid ab.$$

② Step.

(*)1 Identify nullable variable

(*)2 In the production rule

have the productions with & without nullable variable ($\Rightarrow$ R.H.S) & remove the

rule $A \rightarrow \epsilon$

Note: While eliminating the context or meaning of the Grammar should not change.

$$\begin{aligned} 2. \quad & S \rightarrow AB \\ & A \rightarrow aAA | \epsilon \\ & B \rightarrow bBB | \epsilon \end{aligned}$$

Nullable variable: A & B.

$$\begin{aligned} S &\rightarrow AB | A | B | \epsilon \\ A &\rightarrow aAA | aA | a \\ B &\rightarrow bBB | bB | b \end{aligned}$$

$$\begin{aligned} 3. \quad & S \rightarrow AbaC \\ & A \rightarrow BC \\ & B \rightarrow b | \epsilon \\ & C \rightarrow D | \epsilon \\ & D \rightarrow d \end{aligned}$$

Nullable variable: A, B & C

$$\begin{aligned} S &\rightarrow AbaC | Aba | baC | ba \\ A &\rightarrow BC | B | C \\ B &\rightarrow b \\ C &\rightarrow D \\ D &\rightarrow d \end{aligned}$$

$$\begin{aligned} 4. \quad & S \rightarrow aA \\ & A \rightarrow b | \epsilon \end{aligned}$$

Nullable variable: A

$$\begin{aligned} S &\rightarrow aA | a \\ A &\rightarrow b \end{aligned}$$

$$\begin{aligned} 5. \quad & S \rightarrow ABAC \\ & A \rightarrow aA | \epsilon \\ & B \rightarrow bB | \epsilon \\ & C \rightarrow c \end{aligned}$$

Nullable variable: A & B

$$\begin{aligned} S &\rightarrow ABAC | ABC | BAC | AC | \\ &\quad AAC | BC | c \end{aligned}$$

$$\begin{aligned} A &\rightarrow aA | a \\ B &\rightarrow bB | b \\ C &\rightarrow c \end{aligned}$$

$$\begin{aligned} 6. \quad & S \rightarrow aSa \\ & S \rightarrow bSb | \epsilon \end{aligned}$$

Nullable variable: S

$$S \rightarrow aSa | aa | bSb | bb | \epsilon$$

$$\begin{aligned} 7. \quad & S \rightarrow a/xb/aYa \\ & X \rightarrow Y | \epsilon \\ & Y \rightarrow b/X \end{aligned}$$

Nullable variable: X, Y

$$\begin{aligned} S &\rightarrow a/xb/aYa | b/aa \\ X &\rightarrow Y \\ Y &\rightarrow b/X \end{aligned}$$

⑧. Design a CFG for regular expression $x = (a+b)^* bb(a+b)^*$, which is free from ϵ -productions.

$$S \rightarrow AbbA$$

$$A \rightarrow aA | bA | \epsilon$$

Nullable variable: A

$$S \rightarrow AbbA | Abb | bbA | bb$$

$$A \rightarrow aA | bA | a | b$$

Algorithm, (Finding an equivalent CFG with no ϵ -production):

Given a CFG $G = (V, T, P, s)$,

Construct a CFG $G_1 = (V, T, P_1, s)$ with no ϵ -productions as follows.,.

(*)₁. Initialize P_1 to be P .

(*)₂. Find all nullable variables in V ,

(*)₃. For every production $A \rightarrow \alpha$ in P , add to P_1 every production that can be obtained from this one by deleting from α one or more of the occurrences of nullable variables in α .

(*)₄. Delete all ϵ -productions from P_1 . Also delete any duplicates, as well as productions of the form $A \rightarrow A$.

Algorithm 2

construct as follows

(*)₁. Find

(*)₂. For

(*)₃. For

every

$A \rightarrow \alpha$

(*)₄. Delete

①.

$S \rightarrow A$

$B \rightarrow$

$A \rightarrow$

②.

S

A

B

C

D

E

Elimination of Unit Productions:
Algorithm 2: (Finding an equivalent CFG with no unit productions)
 Given a CFG (V, T, P, s) with no ϵ -productions,
 construct a grammar $G_1 = (V, T, P_1, s)$ having no unit productions
 as follows...

- (*) Initialize P_1 to be P .
- (*) For each $A \in V$, find the set of A -derivable variables.
- (*) For every pair (A, B) such that B is A -derivable and every nonunit production $B \rightarrow \alpha$, add the production $A \rightarrow \alpha$ to P_1 if it is not already present in P_1 .
- (*) Delete all unit productions from P_1 .

①. $S \rightarrow Aa \mid B$
 $B \rightarrow A \mid bb$
 $A \rightarrow a \mid bc \mid B$.

$S \rightarrow Aa \mid bb \mid a \mid bc$
 $B \rightarrow bb \mid a \mid bc$
 $A \rightarrow a \mid bc \mid bb$

$B \rightarrow A \rightarrow a \mid bc \mid B$ ($B \rightarrow A$)
 $A \rightarrow B \rightarrow A \mid bb$ ($A \rightarrow B$)
 $S \rightarrow B \rightarrow bb \mid a \mid bc$ ($S \rightarrow B$)

②. $S \rightarrow AB$
 $A \rightarrow a$
 $B \rightarrow C \mid b$
 $C \rightarrow D$
 $D \rightarrow E$
 $E \rightarrow a$

$S \rightarrow AB$
 $A \rightarrow a$
 $B \rightarrow b \mid a$
 $C \rightarrow a$
 $D \rightarrow a$
 $E \rightarrow a$

$B \rightarrow C \rightarrow D \rightarrow E \rightarrow a$ ($B \rightarrow C$)
 $C \rightarrow D \rightarrow E \rightarrow a$ ($C \rightarrow D$)
 $D \rightarrow E \rightarrow a$ ($D \rightarrow E$)

Derivable:

- ①. If $A \rightarrow B$ is a production, B is A -derivable.
- ②. If C is A -derivable, $C \rightarrow B$ is a production, and $B \neq A$, then B is A -derivable.
- ③. No other variables are A -derivable.
- ④. A variable A is A -derivable only if $A \rightarrow A$ is actually a production.

$$\begin{aligned} \textcircled{3}. \quad S &\rightarrow S + T \mid T \\ T &\rightarrow T * F \mid F \\ F &\rightarrow (s) \mid a. \end{aligned}$$

$$\begin{aligned} S &\rightarrow S + T \mid T * F \mid (s) \mid a \\ T &\rightarrow T * F \mid (s) \mid a \\ F &\rightarrow (s) \mid a \end{aligned}$$

$$\begin{aligned} S &\rightarrow T \\ T &\rightarrow T * F \mid (s) \mid a \\ F &\rightarrow (s) \mid a. \end{aligned}$$

$$\begin{aligned} \textcircled{A}. \quad I &\rightarrow a \mid b \mid Ia \mid Ib \mid Io \mid I \\ F &\rightarrow I \mid (E) \\ T &\rightarrow F \mid T * F \\ E &\rightarrow T \mid E + T. \end{aligned}$$

$$\begin{aligned} I &\rightarrow a \mid b \mid Ia \mid Ib \mid Io \mid I \\ F &\rightarrow I \mid (E) \mid a \mid b \mid Ia \mid Ib \mid Io \mid I \\ T &\rightarrow T * F \mid (E) \mid a \mid b \mid Ia \mid Ib \mid Io \mid I \\ E &\rightarrow E + T \mid T * F \mid (E) \mid a \mid b \mid Ia \mid Ib \mid Io \mid I \end{aligned}$$

⑤.

Elimination

two this

④. We u

string w

so can

zero sta

②. We

$S \Rightarrow a$

reachable

①. S

A

B

C

⑤.

$$S \rightarrow A | bb$$

$$A \rightarrow B | b$$

$$B \rightarrow S | a$$

$$S \rightarrow bb | a | b$$

$$A \rightarrow b | bb | a$$

$$B \rightarrow a | bb | b$$

$$S \rightarrow A \rightarrow \textcircled{B} | b$$

$$\downarrow$$

$$\textcircled{S} | a$$

$$A \rightarrow \textcircled{B} | b$$

$$\textcircled{S} | a | b$$

$$\textcircled{A} | bb$$

$$B \rightarrow \textcircled{S} | a$$

$$\downarrow$$

$$\textcircled{A} | bb | a$$

$$\textcircled{B} | b$$

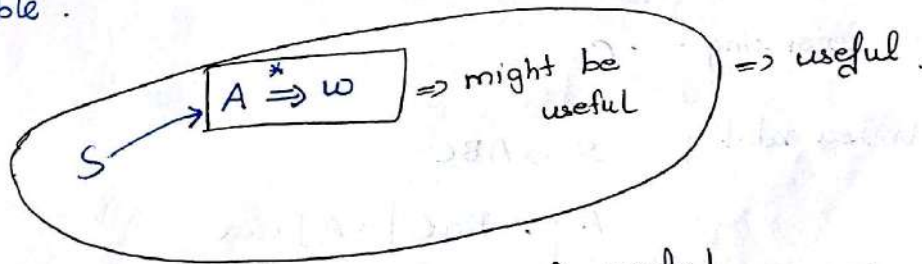
Elimination of Useless Variables:

Elimination of Useless Variables involves

two things...

①. We say X is generating if $X \xRightarrow{*} w$ for some terminal string w . Note that every terminal is generating, since w can be that terminal itself, which is derived by zero steps.

②. We say X is reachable if there is a derivation $S \xRightarrow{*} \alpha X \beta$ for some α and β . Useful symbol will be both generating and reachable.



$$A \xRightarrow{*} aBb \Rightarrow \text{not useful}$$

①. $S \rightarrow AB | a$

$$A \rightarrow BC | b$$

$$B \rightarrow aB | c$$

$$C \rightarrow ac | B$$

Not Generating:
Useless variables: B, C

$$S \rightarrow AB | a$$

$$A \rightarrow BC | b$$

On Reachability: $S \rightarrow AB$ not reachable
 $A \rightarrow BC$ not reachable
 $A \rightarrow b$ no use

$$\boxed{S \rightarrow a}$$

$$\begin{aligned}
 (2) \quad & S \rightarrow AB/AC \\
 & A \rightarrow aAb/bAa/a \\
 & B \rightarrow bbA/aaB/AB \\
 & C \rightarrow abCA/aDb \\
 & D \rightarrow bD/aC
 \end{aligned}$$

Useless Variables:

Not Generating: C, D

Un Reachability: $S \rightarrow AC$

$$\begin{aligned}
 \therefore & \boxed{
 \begin{aligned}
 S &\rightarrow AB \\
 A &\rightarrow aAb/bAa/a \\
 B &\rightarrow bbA/aaB/AB
 \end{aligned}
 }
 \end{aligned}$$

$$\begin{aligned}
 (3) \quad & S \rightarrow ABC/BaB \\
 & A \rightarrow aA/BaC/aaa \\
 & B \rightarrow bBb/a \\
 & C \rightarrow CA/AC
 \end{aligned}$$

Useless Variables:

Not Generating: C

Un Reachability: $S \rightarrow ABC$

$A \rightarrow BaC/aA/aaa$

$$\therefore \boxed{
 \begin{aligned}
 S &\rightarrow BaB \\
 B &\rightarrow bBb/a
 \end{aligned}
 }$$

$$\begin{aligned}
 (4) \quad & S \rightarrow AB/a \\
 & A \rightarrow b
 \end{aligned}$$

Useless variables: Not Generating: B

Un Reachability: $S \rightarrow AB$

$A \rightarrow b$

$$\therefore \boxed{S \rightarrow a}$$

(5).

$S \rightarrow$
 $A \rightarrow$
 $B \rightarrow$
 $X \rightarrow$

Useless variables

(6).

$A \rightarrow$
 $X \rightarrow$
 $Y \rightarrow$
 $Z \rightarrow$

Useless variables

(7).

$S \rightarrow$
 $A \rightarrow$
 $B \rightarrow$
 $C \rightarrow$

Useless variables
 Not Generating
 Un Reachability

5. $S \rightarrow aB | bX$
 $A \rightarrow BAd | bSX | a$
 $B \rightarrow aSB | bBX$
 $X \rightarrow SBD | aBx | ad.$

Useless variables:

Not. Generating: $D, B.$

Unreachable: $S \rightarrow aB$
 $A \rightarrow BAd | bSX | a$
 $B \rightarrow aSB | bBX$
 $X \rightarrow SBD | aBx.$

$$\therefore \begin{cases} S \rightarrow bX \\ X \rightarrow ad \end{cases}$$

6. $A \rightarrow xyz / xyzz$
 $X \rightarrow xz / xYz$
 $Y \rightarrow yYy / xz$
 $Z \rightarrow Zy / z$

Useless variables:

Not. Generating: X, Y

Unreachable: $Z \rightarrow Zy | z$

$$\therefore A \rightarrow xyz$$

7. $S \rightarrow aC / SB$
 $A \rightarrow bSCa$
 $B \rightarrow aSB / bBC$
 $C \rightarrow aBC / ad.$

Useless variables:

Not. Generating: B

Unreachable: $A \rightarrow bSCa$

$$\therefore \begin{cases} S \rightarrow aC \\ C \rightarrow ad \end{cases}$$

a

CHOMSKY NORMAL FORM:

A CFG is in CNF if every production is of one of these two types:

$$A \rightarrow BC$$

$$A \rightarrow a$$

where A, B and C are variables and a is a terminal symbol.

Converting a CFG to CNF:

- ①. Eliminating ϵ -productions.
- ②. Eliminating unit productions.
- ③. Eliminating Useless symbols.

- ④. Restricting the right sides of productions to single terminals or strings of two variables.

①. Convert the below grammar to CNF:

$$S \rightarrow ASB | \epsilon$$

$$A \rightarrow aAS | a$$

$$B \rightarrow SbS | A | bb$$

- ①. Eliminate ϵ -production.,

$$S \rightarrow ASB | AB$$

$$A \rightarrow aAS | a | aA$$

$$B \rightarrow SbS | A | bb | Sb | bS | b$$

- ②. Eliminate unit-production.,

$$S \rightarrow ASB | AB$$

$$A \rightarrow aAS | a | aA$$

$$B \rightarrow SbS | bb | Sb | bS | b | aAS | a | aA$$

- ③. No. Useless symbols.,

④. Restriction

S -

A -

B -

X₁

X₂

X_a

X_b

②. S -

A -

C -

D -

①. Eliminate

②. Eliminate

S -

A -

C -

D -

③. No. Useless

④. Restriction

S -

A -

C -

D -

X₄

④. Restricting the right sides of productions to single terminal or strings of 2 variables.

$$S \rightarrow \cancel{AB} \mid \cancel{X_1 B} \mid AB$$

$$A \rightarrow X_a X_1 \mid X_a \mid X_a A$$

$$B \rightarrow S X_2 \mid X_b X_b \mid S X_b \mid X_b S \mid X_b \mid X_a X_1 \mid X_a \mid X_a A$$

$$X_1 \rightarrow AS$$

$$X_2 \rightarrow X_b S$$

$$X_a \rightarrow a$$

$$X_b \rightarrow b$$

②. $S \rightarrow AACD$

$$A \rightarrow aAb \mid \epsilon$$

$$C \rightarrow aC \mid a$$

$$D \rightarrow aDa \mid bDb \mid \epsilon$$

①. Eliminate ϵ -production.

$$S \rightarrow AACD \mid ACD \mid CD \mid AAC \mid AC \mid C$$

$$A \rightarrow aAb \mid ab$$

$$C \rightarrow aC \mid a$$

$$D \rightarrow aDa \mid bDb \mid aa \mid bb$$

②. Eliminate Unit production.

$$S \rightarrow AACD \mid ACD \mid CD \mid AAC \mid AC \mid aC \mid a$$

$$A \rightarrow aAb \mid ab$$

$$C \rightarrow aC \mid a$$

$$D \rightarrow aDa \mid bDb \mid aa \mid bb.$$

③. No. Useless s/ms.

④. Restricting Right sides of productions to single terminal or strings of 2 variable.

$$S \rightarrow X_2 X_1 \mid A X_1 \mid CD \mid X_2 C \mid AC \mid X_a C \mid a$$

$$A \rightarrow X_3 X_b \mid ab$$

$$C \rightarrow X_a C \mid a$$

$$D \rightarrow X_4 X_a \mid X_5 X_b \mid X_a X_b$$

$$X_2 \rightarrow AA$$

$$X_1 \rightarrow CD$$

$$X_a \rightarrow a$$

$$X_b \rightarrow b$$

$$X_3 \rightarrow X_a A$$

$$X_4 \rightarrow X_a D$$

$$X_5 \rightarrow X_b D$$

OR P.T.O.

Hint:

$$A \rightarrow BCD BCF$$

$$A \rightarrow BY_1$$

$$Y_1 \rightarrow CY_2$$

$$Y_2 \rightarrow DY_3$$

$$Y_3 \rightarrow BY_4$$

$$Y_4 \rightarrow CE$$

④.

$$S \rightarrow AT_1$$

$$S \rightarrow AV_1$$

$$S \rightarrow AV_1$$

$$T_1 \rightarrow AT_2$$

$$V_1 \rightarrow CD$$

$$V_1 \rightarrow AC$$

$$T_2 \rightarrow CD$$

$$S \rightarrow CD | AC | XaC | a$$

$$A \rightarrow XaW_1$$

$$A \rightarrow XaXb$$

$$W_1 \rightarrow AXb$$

$$C \rightarrow XaC | a$$

$$Y_1 \rightarrow DXa$$

$$D \rightarrow xaY_1$$

$$Z_1 \rightarrow DXb$$

$$D \rightarrow XbZ_1$$

$$D \rightarrow XaXb | XbXb$$

$$Xa \rightarrow a$$

$$Xb \rightarrow b$$

③.

$$S \rightarrow bA | aB$$

$$A \rightarrow bAA | aS | a$$

$$B \rightarrow aBB | bS | a$$

CNF

①. No ϵ -production.

②. No unit production.

③. No Useless production.

④. CNF form.

$$S \rightarrow XbA | XaB$$

$$A \rightarrow XbX_1 | XaS | a$$

$$B \rightarrow XaX_2 | XbS | a$$

$$X_1 \rightarrow AA$$

$$X_2 \rightarrow BB$$

$$Xa \rightarrow a$$

$$Xb \rightarrow b$$

$$\begin{aligned} \textcircled{1}. \quad & S \rightarrow 1A \mid 0B \\ & A \rightarrow 1AA \mid 0S \mid 0 \\ & B \rightarrow 0BB \mid 1. \end{aligned}$$

CNF:

- ①. No ϵ -production
- ②. No unit production
- ③. No useless symbol
- ④. CNF Form.,.

$$\begin{aligned} S &\rightarrow X_1 A \mid X_0 B \\ A &\rightarrow X_1 X_2 \mid X_0 S \mid 0 \\ B &\rightarrow X_0 X_3 \mid 1 \\ X_0 &\rightarrow 0 \\ X_1 &\rightarrow 1 \\ X_2 &\rightarrow AA \\ X_3 &\rightarrow BB \end{aligned}$$

$$\begin{aligned} \textcircled{3}. \quad & S \rightarrow a \\ & S \rightarrow b \\ & S \rightarrow \text{csc} \text{ cSS} \end{aligned}$$

CNF:

- ①. No ϵ -production
- ②. No unit production
- ③. No Useless production
- ④. CNF Form.,.

$$\begin{aligned} S &\rightarrow a \\ S &\rightarrow b \\ S &\rightarrow CX \\ X &\rightarrow SS \end{aligned}$$

$$\begin{aligned} \textcircled{6}. \quad & S \rightarrow abSb \mid a \mid aAb \\ & A \rightarrow bS \mid aAAb \end{aligned}$$

CNF:

- ①, ②, ③ no ϵ , unit & useless production.

④. CNF Form.,.

$$\begin{aligned} S &\rightarrow X_1 X_2 \mid a \mid X_a X_4 \\ A &\rightarrow X_b S \mid X_5 X_4 \\ X_a &\rightarrow a \\ X_b &\rightarrow b \\ X_1 &\rightarrow X_a X_b \\ X_2 &\rightarrow SX_b \\ X_3 &\rightarrow X_a X_4 \\ X_4 &\rightarrow AX_b \\ X_5 &\rightarrow X_a A \end{aligned}$$

$$Z = \{a, b, c\}$$

⑦. Design a CFG for the language $L = \{a^{4n} \mid n \geq 1\}$ and convert CFG into CNF form.

$$\begin{aligned} S &\rightarrow AAAAS \mid AAAA \\ A &\rightarrow a \end{aligned}$$

CNF:
①, ②, ③, no ϵ , unit & useless production.

$$\begin{aligned} S &\rightarrow X_1 X_3 \mid X_1 X_1 \\ A &\rightarrow a \\ X_1 &\rightarrow AA \\ X_3 &\rightarrow X_1 S \end{aligned}$$

Greibach normal form (GNF):

The CFG is said to be in GNF if all the productions are of the form...

$$A \rightarrow a\alpha$$

where $a \in T$ and $\alpha \in V^*$

Converting a CFG to GNF:

- (*)₁. Obtain the grammar in CNF.
- (*)₂. Rename the non-terminals to $A_1, A_2, A_3, \dots$
- (*)₃. Using substitution method obtain the productions to the form

$$A_i \rightarrow A_j \alpha \quad \text{for } i < j$$

where $\alpha \in V^*$

Substitution Method:

$$A \rightarrow x, Bx_2$$

$$B \rightarrow y_1 | y_2 | \dots | y_n$$

} P

Then

$$A \rightarrow x, y_1 x_2 | x, y_2 x_2 | \dots | x, y_n x_2 \quad \left. \vphantom{A \rightarrow} \right\} P_1$$

Remove $B \rightarrow y_1 | y_2 | \dots | y_n$

- (*)₄. After substitution, if a grammar has left-recursion, eliminate left recursion...

Left recursion:

$$A \rightarrow A\alpha_1 | A\alpha_2 | A\alpha_3 \dots | A\alpha_n | \beta_1 | \beta_2 | \beta_3 \dots \beta_m \quad \left. \vphantom{A \rightarrow} \right\} P_1$$

After removing left recursion...

$$A \rightarrow \beta_1 A' | \beta_2 A' | \beta_3 A' | \dots | \beta_m A'$$

$$A' \rightarrow \alpha_1 A' | \alpha_2 A' | \alpha_3 A' | \dots | \alpha_n A' | \epsilon$$

Step 3 and step 4 can be repeated for any number of times.

①. Convert the grammar.

$$\begin{aligned} S &\rightarrow AB|BC \\ A &\rightarrow aB|bA|a \\ B &\rightarrow bB|cC|b \\ C &\rightarrow c \end{aligned}$$

into GNF.

$$S \rightarrow AB|BC \quad \text{--- ①}$$

$$A \rightarrow aB|bA|a \quad \text{--- ②}$$

$$B \rightarrow bB|cC|b \quad \text{--- ③}$$

$$C \rightarrow c \quad \text{--- ④}$$

②, ③, & ④ are in GNF.

$$S \rightarrow AB|BC \quad \text{--- ①}$$

replace ① with ② and ③

$$S \rightarrow aBB|bAB|aB|bBC|cCC|bC$$

$$A \rightarrow aB|bA|a$$

$$B \rightarrow bB|cC|b$$

$$C \rightarrow c$$

②. $S \rightarrow abaSa/aba$

Introduce 2 new productions...

$$\begin{aligned} A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

③

∴ GNF

$$S \rightarrow aBASA|aBA$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$\begin{aligned} \textcircled{3} \quad & S \rightarrow AB \\ & A \rightarrow BS \mid a \\ & B \rightarrow SA \mid b \end{aligned}$$

*1 Grammar is in CNF

*2 Rename the variables...

$$\begin{aligned} A_1 &\rightarrow A_2 A_3 \\ A_2 &\rightarrow A_3 A_1 \mid a \\ A_3 &\rightarrow A_1 A_2 \mid b \end{aligned}$$

*3.

$$A_1 \rightarrow A_2 A_3$$

$$A_2 \rightarrow A_3 A_1 \mid a$$

$$A_3 \rightarrow b$$

can be converted to GNF easily.

$$A_3 \rightarrow A_1 A_2$$

Replace A_1 .

$$\therefore A_3 \rightarrow A_2 A_3 A_2$$

Replace right side A_2

$$A_3 \rightarrow A_3 A_1 A_3 A_2 \mid a A_3 A_2 \mid b$$

*4. Remove left recursion

$$A_3 \rightarrow a A_3 A_2 B_3 \mid b B_3 \mid a A_3 A_2 \mid b \quad \textcircled{3}$$

$$B_3 \rightarrow A_1 A_3 A_2 B_3 \mid A_1 A_3 A_2$$

*5. Replace A_3 in A_2

$$\therefore A_2 \rightarrow a A_3 A_2 B_3 A_1 \mid b B_3 A_1 \mid a A_3 A_2 A_1 \mid b A_1 \mid a \quad \textcircled{2}$$

Replace
 $\therefore A_1 \rightarrow$

Replace

$B_3 \rightarrow$

doubt

①, ②,

required GNF

④ $S \rightarrow$

$A \rightarrow$

$B \rightarrow$

*1 $A \& B$

*2 $S \rightarrow$

$S \rightarrow$

$A \rightarrow$

$B \rightarrow$

⑤ $S \rightarrow$

S

A

B

(b). Replace A_2 in A_1 .

$$\therefore A_1 \rightarrow a A_3 A_2 B_3 A_1 A_3 \mid b B_3 A_1 A_3 \mid a A_3 A_2 A_1 A_3 \mid b A_1 A_3 \mid a A_3 \quad \text{--- (1)}$$

(c). Replace A_1 in B_3 .

$$\begin{aligned} B_3 \rightarrow & a A_3 A_2 B_3 A_1 A_3 A_2 B_3 \mid b B_3 A_1 A_3 A_3 A_2 B_3 \mid \\ & a A_3 A_2 A_1 A_3 A_3 A_2 B_3 \mid b A_1 A_3 A_3 A_2 B_3 \mid \\ & a A_3 A_3 A_2 B_3 \mid a A_3 A_2 B_3 A_1 A_3 A_3 A_2 \mid \\ & b B_3 A_1 A_3 A_3 A_2 \mid a A_3 A_2 A_1 A_3 A_3 A_2 \mid \\ & b A_1 A_3 A_3 A_2 \mid a A_3 A_3 A_2 \quad \text{--- (4)} \end{aligned}$$

doubt

(1), (2), (3) & (4) forms the production rules of the required GNF.

(4)

$$\begin{aligned} S &\rightarrow AB \\ A &\rightarrow aA \mid bB \mid b \\ B &\rightarrow b \end{aligned}$$

(*) $A \& B$ is in GNF

(*)

$$\begin{aligned} S &\rightarrow AB \\ S &\rightarrow aAB \mid bBB \mid bB \\ A &\rightarrow aA \mid bB \mid b \\ B &\rightarrow b \quad \text{is in GNF.} \end{aligned}$$

(3)

$$S \rightarrow absb \mid aa$$

$$S \rightarrow aBSB \mid aA$$

$$A \rightarrow a$$

$$B \rightarrow b$$

(2)

⑥. $S \rightarrow aAS$
 $\downarrow$
 doubt. $S \rightarrow a$
 $A \rightarrow SbA$
 $B \rightarrow SS$
 $A \rightarrow ba$

$S \rightarrow aAS$
 $S \rightarrow a$
 $A \rightarrow bA_1$
 $A_1 \rightarrow a$
 $B_1 \rightarrow b$
 $A \rightarrow aASB_1A$
 $A \rightarrow aB_1A$
 $B \rightarrow aASS$
 $B \rightarrow aS$

$A \rightarrow SbA$
 $\rightarrow aASbA$
 $\rightarrow aASB_1A$
 $A \rightarrow SbA$
 $\rightarrow aB_1A$

$B \rightarrow SS$
 $\rightarrow aASS$

⑦. $S \rightarrow AA|0$
 $A \rightarrow SS|1$

*1. Grammar is in CNF.

*2. Rename the variables & Substitution

$A_1 \rightarrow A_2A_2|0$
 $A_2 \rightarrow A_1A_1|1$

$A_2 \rightarrow A_2A_2A_1|0A_1|1$

*3. Remove left recursion.

$A_2 \rightarrow \epsilon A_1|1|0A_1B|1B$ — ②

$B \rightarrow A_2A_1|A_2A_1B$

(*) Sub A_2 in A_1

$$A_1 \rightarrow 0A_1A_2 \mid 1A_2 \mid 0A_1BA_2 \mid 1BA_2 \mid 0 \quad \text{--- (1)}$$

(*) Sub A_2 in B

$$B \rightarrow 0A_1A_1 \mid 1A_1 \mid 0A_1BA_1 \mid 1BA_1 \mid \\ 0A_1A_1B \mid 1A_1B \mid 0A_1BA_1B \mid 1BA_1B \quad \text{--- (3)}$$

(1), (2) and (3) is in GNF.

(8). Convert the following grammar to GNF.

$$E \rightarrow E * T \mid T$$

$$T \rightarrow F + T \mid F$$

$$F \rightarrow a \mid b \mid (E).$$

Push Down Automata (PDA): (FA + Stack)

PDA is defined by $(Q, \Sigma, \delta, q_0, Z_0, F, \Gamma)$

Q : Finite set of states.

Σ : input symbol.

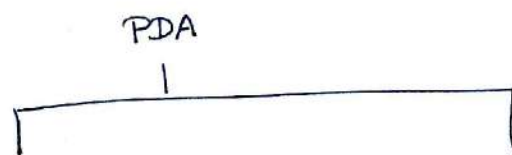
δ : transition function.

q_0 : Initial state.

Z_0 : Bottom of the stack.

F : set of final states. $\Rightarrow$ without final states also we can design PDA.

Γ : stack alphabet.



Deterministic PDA

Non-deterministic PDA

$$\delta: Q \times \{\Sigma \cup \epsilon\} \times \Gamma \rightarrow Q \times \Gamma^*$$

$$\delta: Q \times \{\Sigma \cup \epsilon\} \times \Gamma \rightarrow 2^{(Q \times \Gamma^*)}$$

Instantaneous Description (ID),

During execution, PDA goes through a sequence of configurations. Each configuration is called instantaneous description (ID) consists of (i). state (ii). input yet to be scanned and (iii) stack content & is represented by a triplet (q, x, γ) where q is the current state, x is the input yet to be read (leftmost symbol of x is the current i/p) and γ is the stack content (leftmost symbol of γ is the current top-of-stack symbol).

Move:

The sym

(q, γ) .

and A moves.

Language

@. Acceptance

if $(q_0$

state L

⑤. Acceptance

PDA by

empty st

Note:

Move:

Each move involves a change from one ID to another.
The symbol $\vdash$ is used to represent a move.

$(p, ax, A\alpha) \vdash (q, x, \beta\alpha)$ if $\delta(p, a, A)$ includes (q, β) . [The symbol a is consumed. PDA changes to state q and A is replaced by β]. $\vdash^*$ represents a sequence of moves.

Language Accepted by a PDA:

PDA can accept a string in 2 ways.

① Acceptance by Final State:

An input w is said to be accepted by final state

iff $(q_0, w, z_0) \vdash^* (p, \epsilon, \beta)$ for some p in F .

Language accepted by a PDA M by final state $L(M)$ is defined as:

$$L(M) = \{w \mid (q_0, w, z_0) \vdash^* (p, \epsilon, \beta) \text{ for some } p \text{ in } F\}$$

② Acceptance by Empty Stack:

An input w is said to be accepted by a

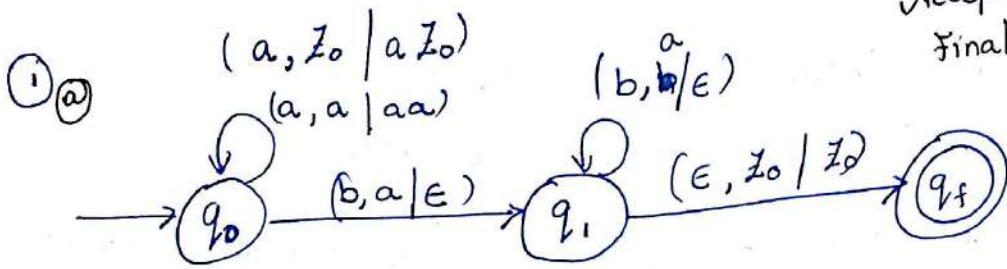
PDA by empty stack iff $(q_0, w, z_0) \vdash^* (p, \epsilon, \epsilon)$

Language accepted by a PDA M by empty stack $N(M)$ is defined as:

$$N(M) = \{w \mid (q_0, w, z_0) \vdash^* (q, \epsilon, \epsilon)\}.$$

Note: a & b are equivalent in power.

Acceptance by Final state.



$\left. \begin{matrix} a, Z_0 | a Z_0 \\ a, a | aa \end{matrix} \right\} \Rightarrow \text{Pushing the elements}$

$\left. \begin{matrix} b, a | \epsilon \\ b, b | \epsilon \end{matrix} \right\} \Rightarrow \text{Popping}$

$\epsilon, Z_0 | Z_0 \Rightarrow$
 $\downarrow$ end of string $\downarrow$ top of stack

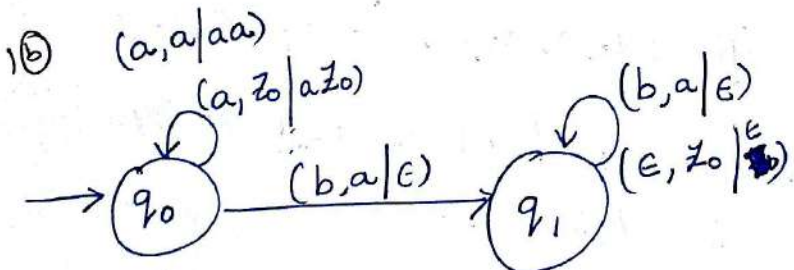
Equivalent representation of the automata:

Acceptance by Final state

$$\begin{aligned}
 \delta(q_0, a, Z_0) &= (q_0, a Z_0) \\
 \delta(q_0, a, aa) &= (q_0, aa) \\
 \delta(q_0, b, a) &= (q_1, \epsilon) \\
 \delta(q_1, b, a) &= (q_1, \epsilon) \\
 \delta(q_1, \epsilon, Z_0) &= (q_f, Z_0)
 \end{aligned}$$

Acceptance by Empty stack

$$\begin{aligned}
 \delta(q_0, a, Z_0) &= (q_0, a Z_0) \\
 \delta(q_0, a, aa) &= (q_0, aa) \\
 \delta(q_0, b, a) &= (q_1, \epsilon) \\
 \delta(q_1, b, a) &= (q_1, \epsilon) \\
 \delta(q_1, \epsilon, Z_0) &= (q_1, \epsilon)
 \end{aligned}$$



Acceptance by Empty stack.

by

Instantaneous Description:

(i) aabb

using (1a)

$(q_0, aabb, z_0) \vdash (q_0, abb, az_0)$

$\vdash (q_0, bb, aa z_0)$

$\vdash (q_1, b, az_0)$

~~$\vdash (q_1, \epsilon, z_0)$~~ $\vdash (q_f, \epsilon, z_0)$

~~$\vdash (q_1, \epsilon, \epsilon)$~~ String accepted

(ii) aabb using (1b)

$(q_0, aabb, z_0) \vdash (q_0, abb, az_0)$

$\vdash (q_0, bb, aa z_0)$

$\vdash (q_1, b, az_0)$

$\vdash (q_1, \epsilon, z_0)$

$\vdash (q_1, \epsilon, \epsilon)$

String accepted

(iii) aabba using (1a)

$(q_0, aabba, z_0) \vdash (q_0, abba, az_0)$

$\vdash (q_0, bba, aa z_0)$

$\vdash (q_1, ba, az_0)$

$\vdash (q_1, a, z_0)$ No next move.

(iv) aabba using (1b)

$(q_0, aabba, z_0) \vdash (q_0, abba, az_0)$

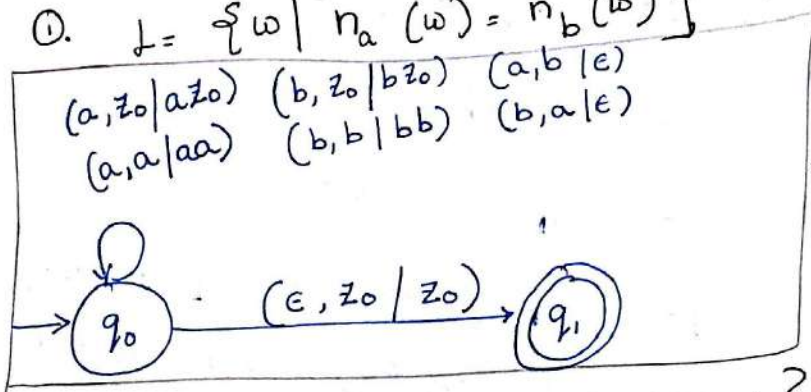
$\vdash (q_0, bba, aa z_0)$

$\vdash (q_1, ba, az_0)$

$\vdash (q_1, a, z_0)$ No next move.

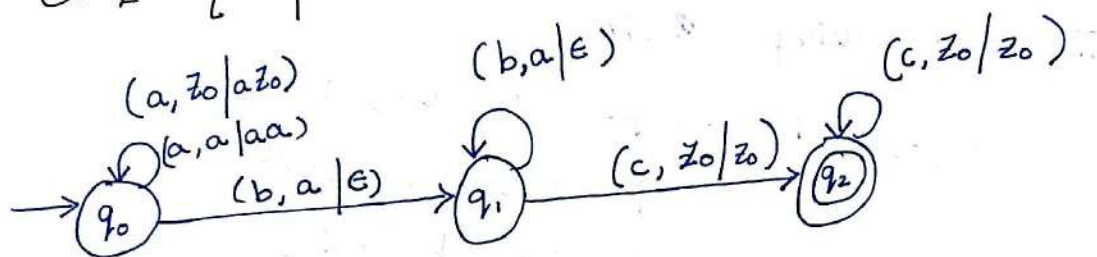
①. Design DPDA for the foll. languages.

①. $L = \{w \mid n_a(w) = n_b(w)\}$

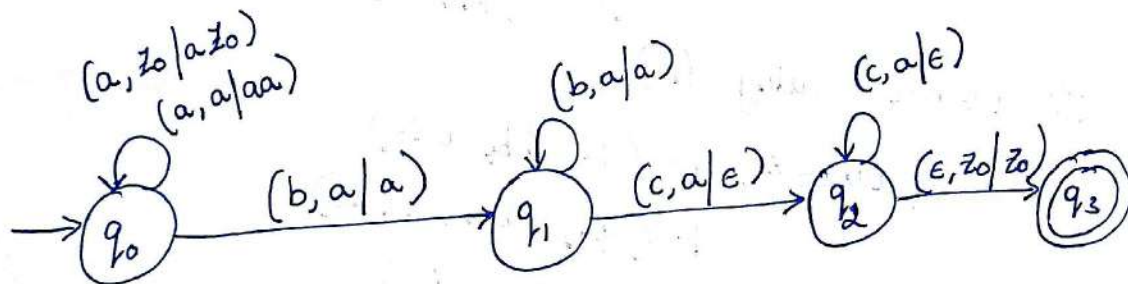


⇒ Non-deterministic PDA.

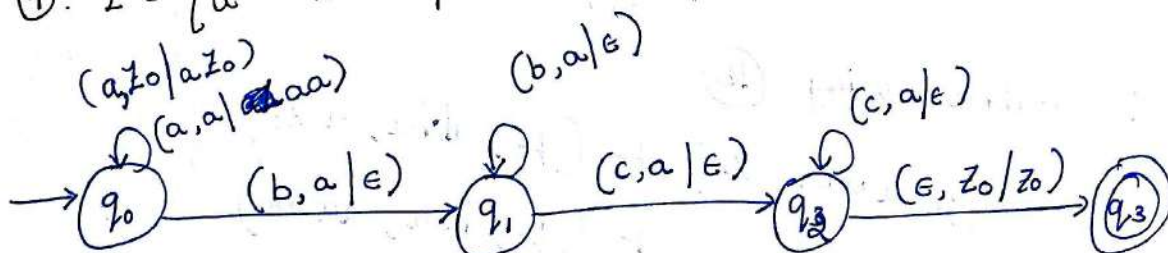
②. $L = \{w \mid a^n b^n c^m, n, m \geq 1\}$



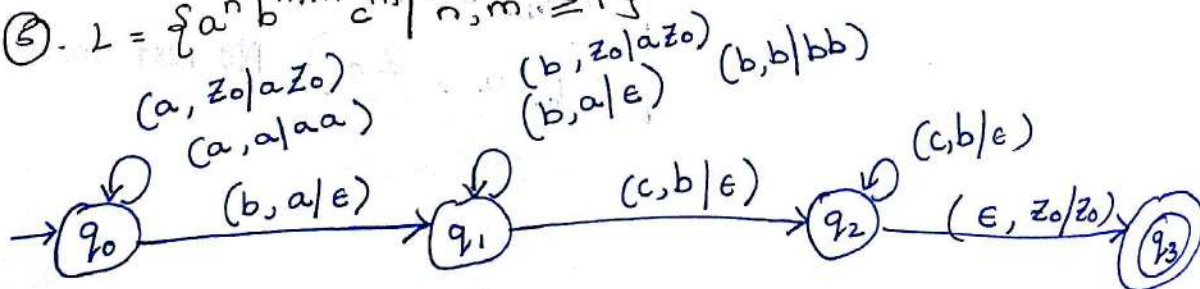
③. $L = \{a^n b^m c^n \mid n, m \geq 1\}$



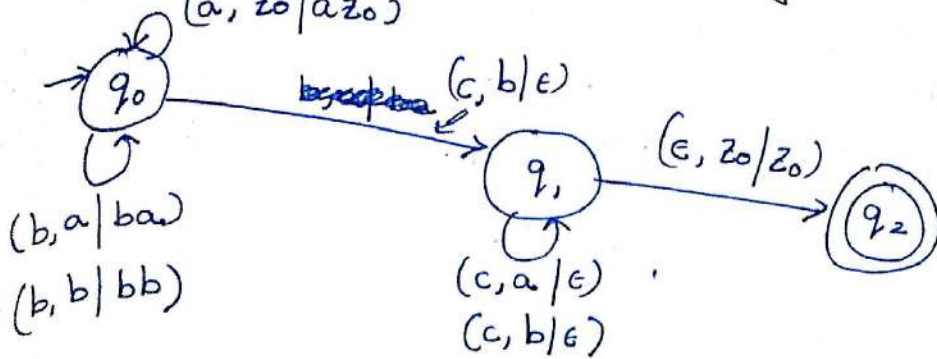
④. $L = \{a^{m+n} b^m c^n \mid m, n \geq 1\}$



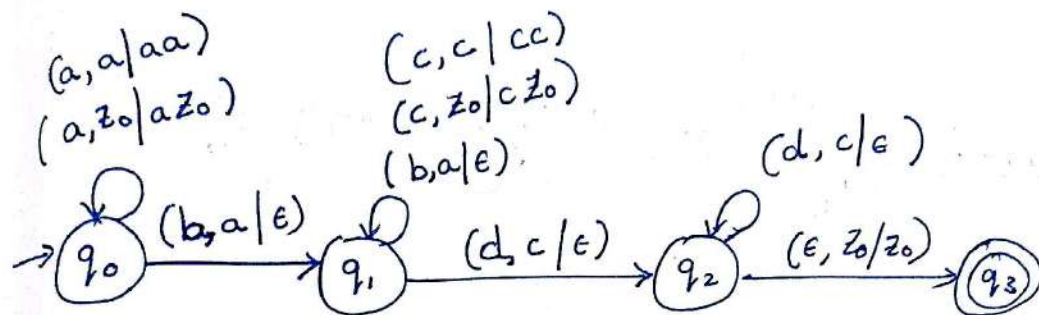
⑤. $L = \{a^n b^{mn} c^m \mid n, m \geq 1\}$



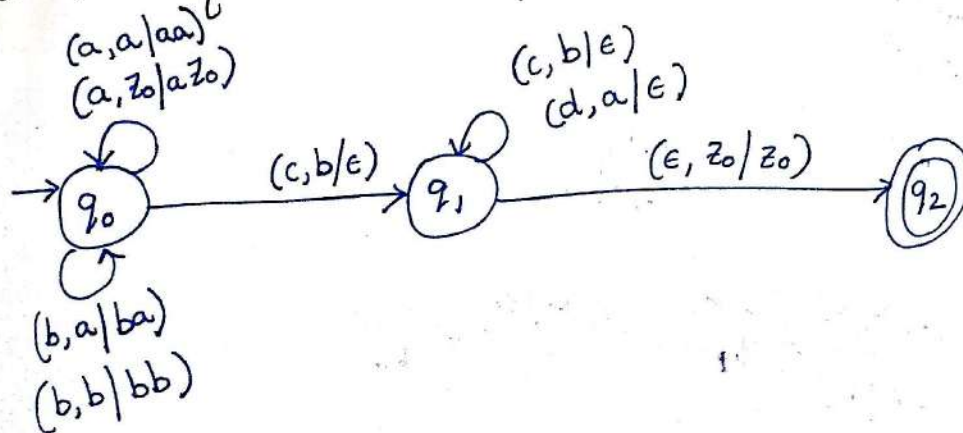
6. $L = \{a^n b^m c^{n+m} \mid n, m \geq 1\}$
 $(a, a \mid aa)$
 $(a, z_0 \mid az_0)$



7. $L = \{a^n b^n c^m d^m \mid n, m \geq 1\}$

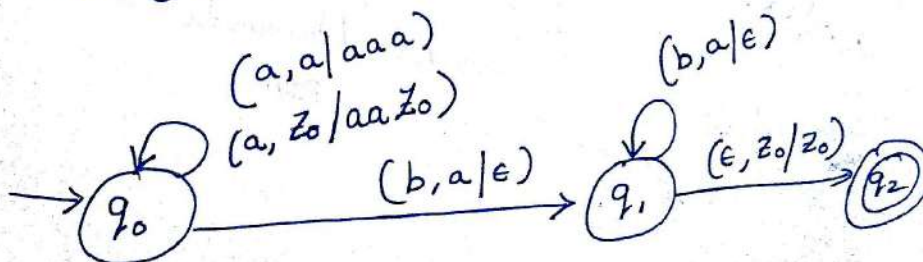


8. $L = \{a^n b^m c^m d^n \mid n, m \geq 1\}$

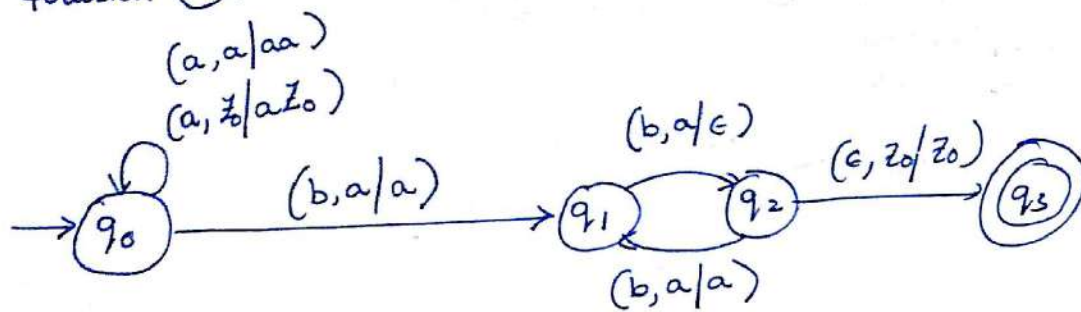


9. $a^n b^{2n} \mid n \geq 1$

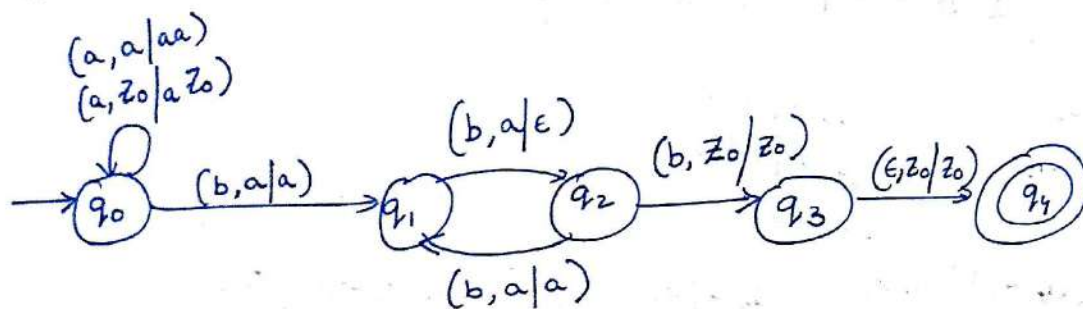
Solution ①:



Solution (2):



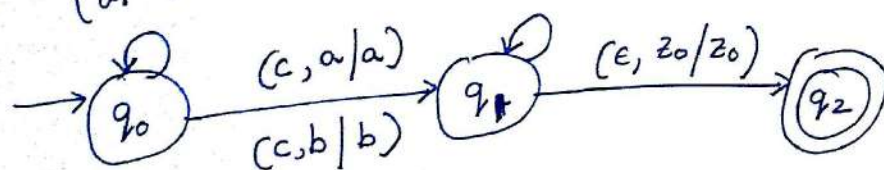
(10). $a^n b^{2n+1}$



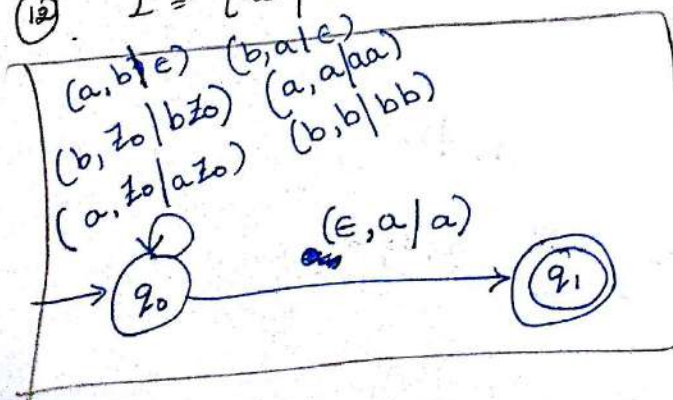
(11). $wcw^R / w \in (a,b)^+$

$(b, b/bb)$
 $(a, a/aa)$
 $(b, z_0/bz_0)$
 $(a, z_0/az_0)$

$(b, b/ε)$
 $(a, a/ε)$

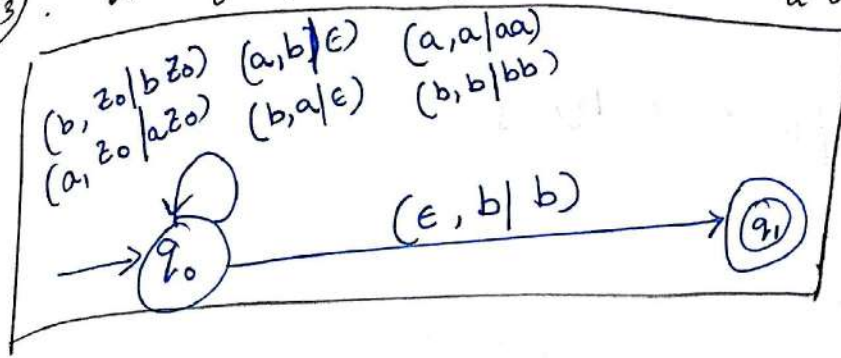


(12). $L = \{w \mid w \in (a,b)^+ \text{ \& \; } n_a(w) > n_b(w)\}$



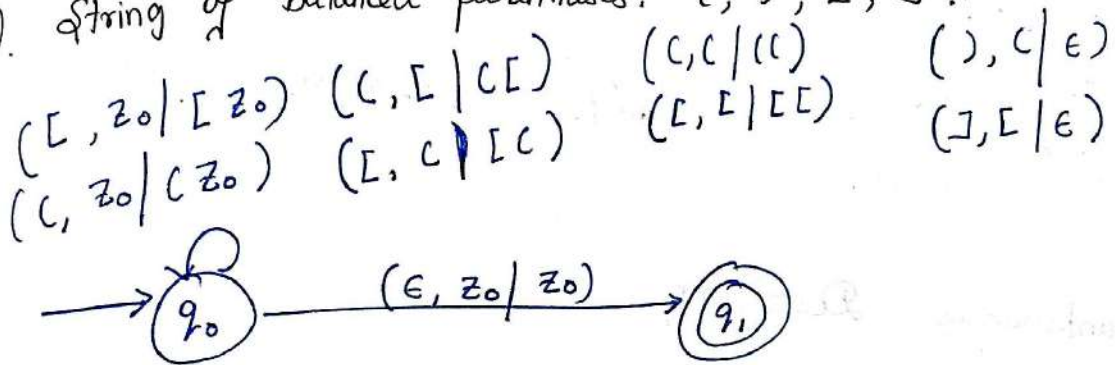
$\Rightarrow$ Non-deterministic PDA.

13. $L = \{w \mid w \in (a,b)^* \text{ \& } n_a(w) \leq n_b(w)\}$



⇒ Non-deterministic PDA

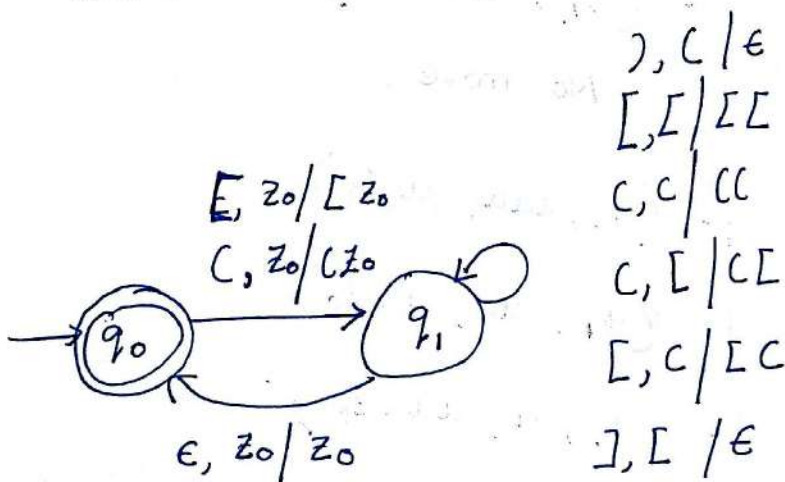
14. String of balanced parentheses. (,), [,].



15. String of balanced parentheses.

Valid strings: ϵ , $[()()([)])$, $[[[]]]()$

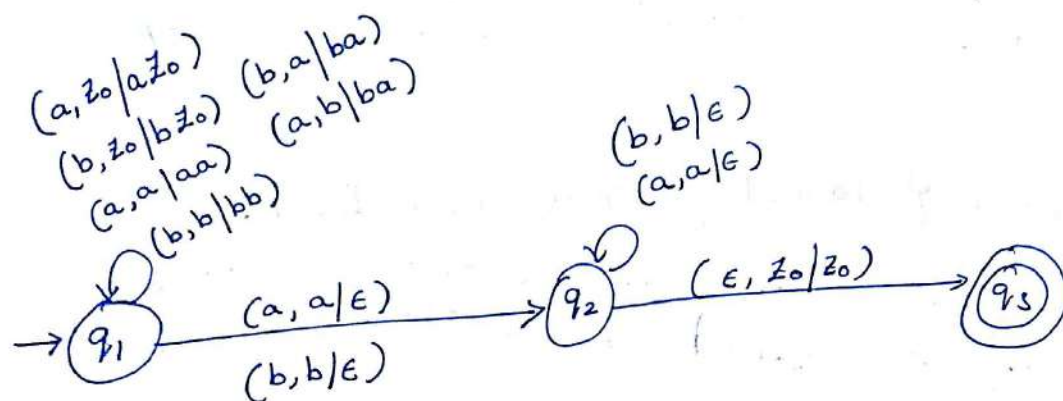
Invalid strings: $[)()[]]$, $)([, [)$



$] , [| \epsilon$
 $[, [| [[$
 $(, z_0 | (z_0$
 $(, [| ([$
 $[,) | \epsilon$
 $[,] | []$
 $] , [| \epsilon$

②. Design $NPDPA^{(0)}$ ^{DPDA} for the foll. languages.

①. $L = \{ ww^R \mid w \in (a,b)^+ \}$.



Instantaneous Description:

aaaa
↓

$$(q_1, aaaa, z_0) \vdash (q_1, aaa, a z_0)$$

$$\vdash (q_2, aa, z_0)$$

No move.

$$(q_1, aaaa, z_0) \vdash (q_1, aaa, a z_0)$$

$$\vdash (q_1, aa, aa z_0)$$

$$\vdash (q_1, a, aaa z_0)$$

$$\vdash (q_1, \epsilon, aaaa z_0)$$

No move.

$$(q_1, aaaa, z_0) \vdash (q_1, aaa, a z_0)$$

$$\vdash (q_1, aa, aa z_0)$$

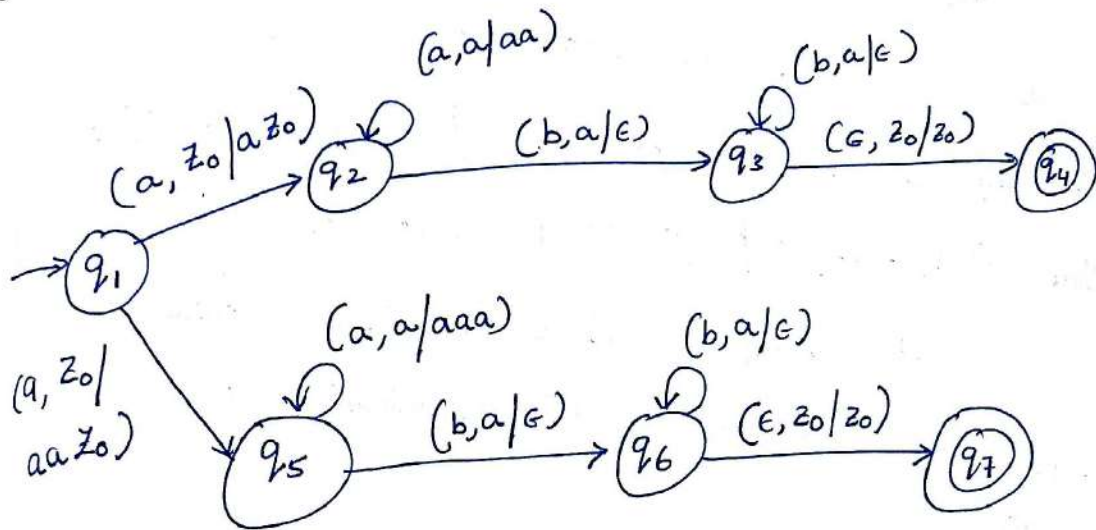
$$\vdash (q_1, a, aaa z_0)$$

$$\vdash (q_1, \epsilon, aa z_0)$$

No move

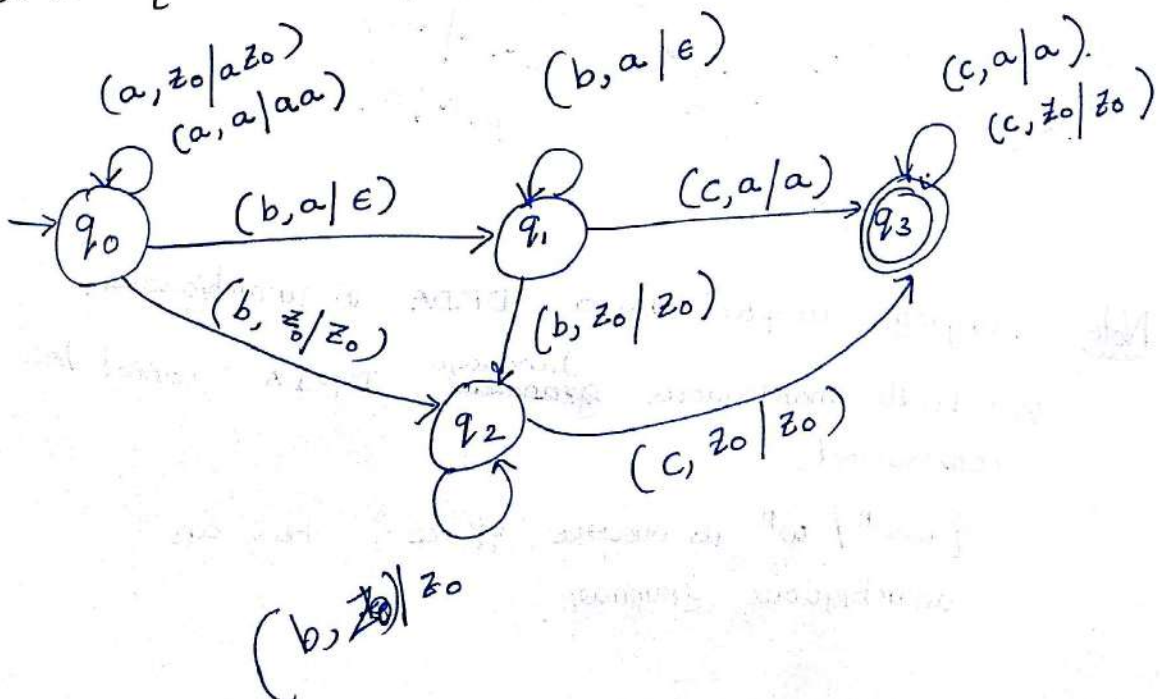
$(q_1, aaaa, z_0) \vdash (q_1, aaa, az_0)$
 $\vdash (q_1, aa, aa z_0)$
 $\vdash (q_2, a, az_0)$
 $\vdash (q_2, \epsilon, z_0)$ String Accepted.

2. $L = \{a^n b^n \mid n \geq 1\} \cup \{a^n b^{2n} \mid n \geq 1\}$



1. $L = \{w \mid w \in \{a, b\}^* \text{ and } \dots\}$

3. $L = \{a^i b^j c^k \mid i \neq j\} \quad i \geq 0, j \geq 1, k \geq 1.$



Deterministic PDA:

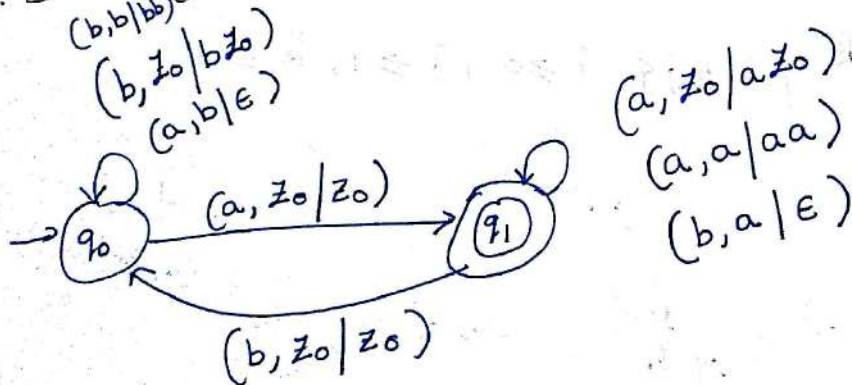
Let $M = (Q, \Sigma, \delta, q_0, F, Z_0, \Gamma)$ M is deterministic if there is no configuration for which M has a choice of more than one move.

In other words, M is deterministic if it satisfies both the following conditions..

- (i). For any $q \in Q$, $a \in \Sigma \cup \{\epsilon\}$ and $X \in \Gamma$, the set $\delta(q, a, X)$ has at most one element.
- (ii). For any $q \in Q$ and $X \in \Gamma$, if $\delta(q, \epsilon, X) \neq \emptyset$, then $\delta(q, a, X) = \emptyset$ for every $a \in \Sigma$.

A language L is a deterministic context-free language (DCFL) if there is a deterministic PDA (DPDA) accepting L .

Q. DPDA for $L = \{x \in \{a, b\}^* \mid n_a(x) > n_b(x)\}$.

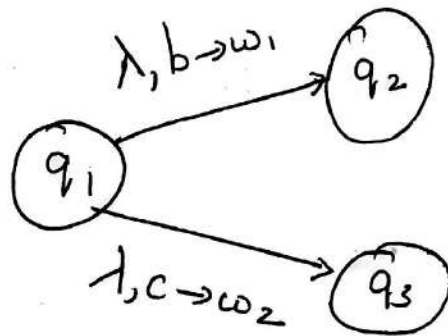
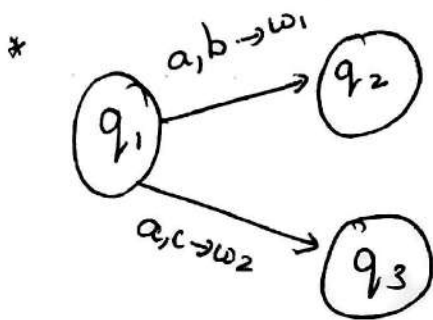
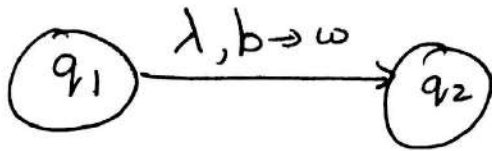
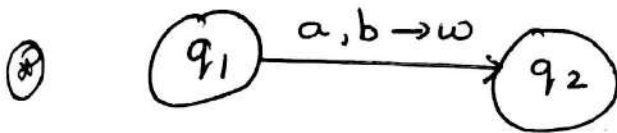


Note: Language accepted by a DPDA is unambiguous.

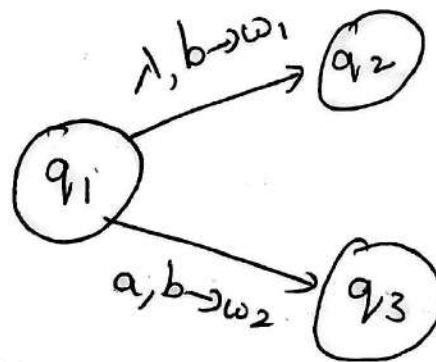
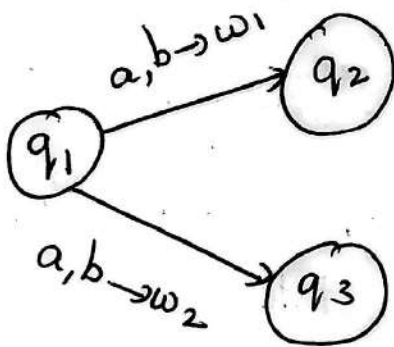
But for all unambiguous ~~language~~ DPDA cannot be constructed.

$\{ww^R \mid w^R \text{ is reverse of } w\}$ has an unambiguous grammar.

Allowed Transitions in DPDA:



Not-Allowed Transitions in DPDA:



CFG to PDA:

Conversion of CFG to PDA

①. Convert the grammar into GNF.

②. For $q_0 \rightarrow$ start state

$z_0 \rightarrow$ top of stack

without consuming any i/p, push the start symbol s onto the stack and change the state to q_1 . The transition for this can be

$$\delta(q_0, \epsilon, z_0) = (q_1, S z_0)$$

③. For each production of the form

$$A \rightarrow a \alpha$$

introduce the transition

$$\delta(q_1, a, A) = (q_1, \alpha)$$

$$A \rightarrow a$$

introduce the transition

$$\delta(q_1, a, A) = (q_1, \epsilon)$$

④. At state q_1 , without consuming any i/p, change the state to q_f which is an accepting state.

The transition will be

$$\delta(q_1, \epsilon, z_0) = (q_f, z_0)$$

①. $S \rightarrow aABC$

$$A \rightarrow aB/a$$

$$B \rightarrow bA/b$$

$$C \rightarrow a$$

①. $\delta(q_0, \epsilon, z_0) = (q_1, S z_0)$

②. $\delta(q_1, a, S) = (q_1, ABC)$

③. $\delta(q_1, a, A) = (q_1, B)$

④. $\delta(q_1, a, A) = (q_1, \epsilon)$

aba

From q_0

$(q_0,$

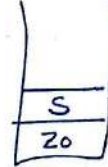
②. $L =$

$M =$

- ⑤. $\delta(q_1, b, B) = (q_1, A)$
 ⑥. $\delta(q_1, b, B) = (q_1, \epsilon)$
 ⑦. $\delta(q_1, a, C) = (q_1, \epsilon)$
 ⑧. $\delta(q_1, \epsilon, z_0) = (q_f, z_0)$

aaba

$S \Rightarrow aABC$
 $\Rightarrow aaBC$
 $\Rightarrow aabc$
 $\Rightarrow aaba$



From PDA

$(q_0, aaba, z_0) \vdash (q_1, aaba, Sz_0)$
 $\vdash (q_1, aba, ABCz_0)$
 $\vdash (q_1, ba, BCz_0)$
 $\vdash (q_1, a, Cz_0)$
 $\vdash (q_1, \epsilon, z_0)$
 $\vdash (q_f, \epsilon, z_0)$

②. $L = \{a^n b^n \mid n \geq 1\}$

$S \rightarrow aSb$
 $S \rightarrow ab$

$S \rightarrow aSB$
 $S \rightarrow aB$
 $B \rightarrow b$

$\delta(q_0, \epsilon, z_0) = (q_1, Sz_0)$

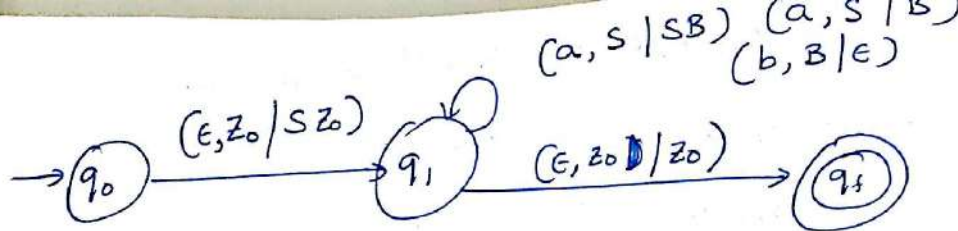
$\delta(q_1, a, S) = (q_1, SB)$

$\delta(q_1, a, S) = (q_1, B)$

$\delta(q_1, b, B) = (q_1, \epsilon)$

$\delta(q_1, \epsilon, z_0) = (q_f, z_0)$

$M = (\{q_0, q_1, q_f\}, \{a, b\}, \delta, q_0, q_f, z_0, \{S, A, B, z_0\})$



- ③. $S \rightarrow oAA$
 $A \rightarrow oS | iS | o.$

$$\delta(q_0, \epsilon, Z_0) = (q_1, SZ_0)$$

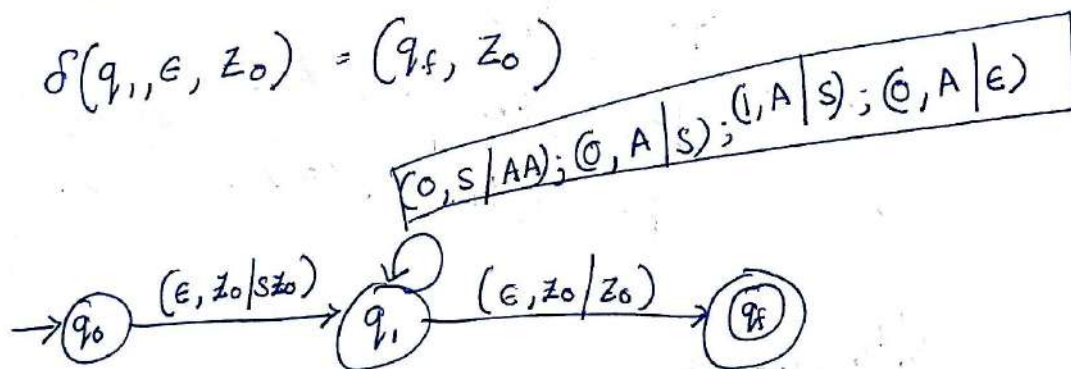
$$\delta(q_1, o, S) = (q_1, AA)$$

$$\delta(q_1, o, A) = (q_1, S)$$

$$\delta(q_1, i, A) = (q_1, S)$$

$$\delta(q_1, o, A) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, Z_0) = (q_f, Z_0)$$



$o10000 \in L(G).$

$$\begin{aligned}
 (q_0, o10000, Z_0) &\vdash (q_1, o10000, SZ_0) \\
 &\vdash (q_1, 10000, AA Z_0) && \vdash (q_1, \epsilon, Z_0) \\
 &\vdash (q_1, 0000, SA Z_0) && \vdash (q_f, \epsilon, Z_0) \\
 &\vdash (q_1, 000, AAA Z_0) \\
 &\vdash (q_1, 00, AA Z_0) \\
 &\vdash (q_1, o, A Z_0)
 \end{aligned}$$

String Accepted.

CFG to

P =

P =

This type empty stack

Conversion

①. For variable

②. Terminal

①. I

F

①. For var

$\delta(q,$

$\delta(q,$

②. Terminal

$\delta(q, a,$

$\delta(q, b,$

$\delta(q, o,$

$\delta(q, i,$

CFG to PDA: (without considering GNF)

$$P = (Q, \Sigma, \delta, q_0, F, Z_0, \Gamma)$$

$$P = (\{q\}, \Sigma, \delta, q, \phi, S, VUT)$$

↓ No final state.

This type of conversion will design PDA with empty stack.

Conversion from CFG to PDA:

①. For variable A :

$$\delta(q, \epsilon, A) = \{ (q, \beta) \mid A \rightarrow \beta \text{ is a production of } P \}$$

②. Terminal a :

$$\delta(q, a, a) = \{q, \epsilon\}.$$

$$\textcircled{1}. \quad \underline{I} \rightarrow a \mid b \mid Ia \mid Ib \mid Io \mid I$$

$$\underline{E} \rightarrow \underline{I} \mid E * E \mid E + E \mid (E).$$

①. For variable E & I

$$\delta(q, \epsilon, E) = \{ (q, E * E), (q, E + E), (q, (E)), (q, I) \}$$

$$\delta(q, \epsilon, I) = \{ (q, a), (q, b), (q, Ia), (q, Ib), (q, Io), (q, I) \}$$

②. Terminal $a, b, o, i, *, +, (,)$

$$\delta(q, a, a) = \{q, \epsilon\}$$

$$\delta(q, b, b) = \{q, \epsilon\}$$

$$\delta(q, o, o) = \{q, \epsilon\}$$

$$\delta(q, i, i) = \{q, \epsilon\}$$

$$\delta(q, *, *) = \{q, \epsilon\}$$

$$\delta(q, +, +) = \{q, \epsilon\}$$

$$\delta(q, (, () = \{q, \epsilon\}$$

$$\delta(q,),) = \{q, \epsilon\}$$

$$\textcircled{2}. \quad S \rightarrow 0S1 \mid A$$

$$A \rightarrow 1A0 \mid S \mid \epsilon$$

Variable S & A :

$$\delta(q, \epsilon, S) = \{ (q, 0S1), (q, A) \}$$

$$\delta(q, \epsilon, A) = \{ (q, 1A0), (q, S), (q, \epsilon) \}$$

Variable 0 & 1 :

$$\delta(q, 0, 0) = \{ q, \epsilon \}$$

$$\delta(q, 1, 1) = \{ q, \epsilon \}$$

Tracing: 0101

$$\delta(q, 0101, S) \vdash \delta(q, 0101, 0S1)$$

$$\vdash \delta(q, \bullet 0101, \overset{S}{\bullet} A1)$$

$$\vdash \delta(q, \bullet 0101, A1)$$

$$\vdash \delta(q, 101, 1A01)$$

$$\vdash \delta(q, 01, A01)$$

$$\vdash \delta(q, 01, 01)$$

$$\vdash \delta(q, 1, 1)$$

$$\vdash \delta(q, \epsilon, \epsilon).$$

String is accepted.

Construction of CFG Equivalent of a PDA:

Let $P = (Q, \Sigma, \delta, q_0, \phi, Z_0, \Gamma)$ be a PDA that accepts a language L by empty stack.

CFG $G = (V, T, P, s)$ that generates L can be constructed using the following rules:

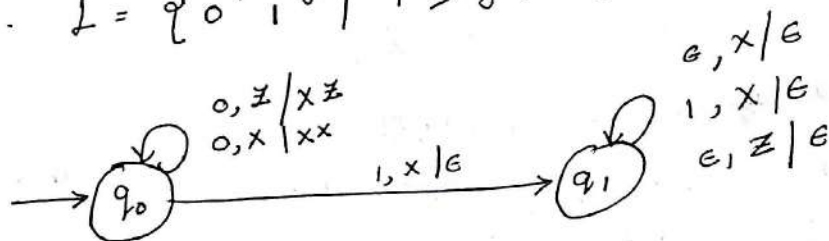
①. For every $q \in Q$, add a production $s \rightarrow [q_0 Z q]$ in P . Thus, if there are n states in a PDA P , then there will be ' n ' new productions added in P .

②. For every $q, r \in Q$, $a \in \{\Sigma \cup \epsilon\}$, $x \in \Gamma$, if $\delta(q, a, x) = (r, \epsilon)$ then add a production $[q x r] \rightarrow a$ in P .

③. For every $q, r \in Q$, $a \in \{\Sigma \cup \epsilon\}$, $x \in \Gamma$ and $x \geq 1$, if $\delta(q, a, x) = (r, x_1, x_2, \dots, x_k)$ where $x_1, x_2, \dots, x_k \in \Gamma$, then for every choice of $q_1, q_2, \dots, q_k \in Q$ add the production

$$[q x q_k] \rightarrow a [r x_1 q_1] [q_1 x_2 q_2] \dots [q_{k-1} x_k q_k] \text{ in } P.$$

$$①. L = \{0^i 1^j \mid i \geq j \geq 1\}$$



Using Rule ①:

$$S \rightarrow [q_0 \neq q_0] \mid [q_0 \neq q_1]$$

Using Rule ②:

$$\delta(q_0, 1, x) = (q_1, \epsilon) \Rightarrow [q_0 \times q_1] \rightarrow 1$$

$$\delta(q_1, \epsilon, x) = (q_1, \epsilon) \Rightarrow [q_1 \times q_1] \rightarrow \epsilon$$

$$\delta(q_1, 1, x) = (q_1, \epsilon) \Rightarrow [q_1 \times q_1] \rightarrow 1$$

$$\delta(q_1, \epsilon, z) = (q_1, \epsilon) \Rightarrow [q_1 \neq q_1] \rightarrow \epsilon$$

Using Rule ③:

$$\delta(q_0, 0, z) = (q_0, xz)$$

①

$$[q_0 \neq q_0] \rightarrow 0 [q_0 \times q_0] [q_0 \neq q_0]$$

$$\rightarrow 0 [q_0 \times q_1] [q_1 \neq q_0]$$

$$[q_0 \neq q_1] \rightarrow 0 [q_0 \times q_1] [q_1 \neq q_1]$$

$$\rightarrow 0 [q_0 \times q_0] [q_0 \neq q_1]$$

$$② \quad \delta(q_0, 0, x) = (q_0, xx)$$

$$[q_0 \times q_0] \rightarrow 0 [q_0 \times q_0] [q_0 \times q_0]$$

$$\rightarrow 0 [q_0 \times q_1] [q_1 \times q_0]$$

$$[q_0 \times q_1] \rightarrow 0 [q_0 \times q_1] [q_1 \times q_1]$$

$$\rightarrow 0 [q_0 \times q_0] [q_0 \times q_1]$$

The required Grammar $G = (V, T, P, S)$ will be

$$V = \{ S, [q_0 \neq q_0], [q_0 \neq q_1], [q_1 \neq q_0], [q_1 \neq q_1], [q_0 \times q_0], [q_0 \times q_1], [q_1 \times q_0], [q_1 \times q_1] \}$$

$$T = \{ 0, 1 \}$$

$$P = \{ S \rightarrow [q_0 \neq q_1] \mid [q_0 \neq q_0] \quad \text{--- (2)}$$

$$[q_0 \times q_1] \rightarrow 1 \quad \text{--- (3)}$$

$$[q_1 \times q_1] \rightarrow 1 \quad \text{--- (4)}$$

$$[q_1 \times q_0] \rightarrow \epsilon \quad \text{--- (5)}$$

$$[q_1 \neq q_1] \rightarrow \epsilon \quad \text{--- (6)}$$

$$[q_0 \neq q_0] \rightarrow 0 [q_0 \times q_0] [q_0 \neq q_0] \mid \quad \text{--- (7)}$$

$$\rightarrow 0 [q_0 \times q_1] [q_1 \neq q_0] \quad \text{--- (8)}$$

$$[q_0 \neq q_1] \rightarrow 0 [q_0 \times q_1] [q_1 \neq q_1] \mid 0 [q_0 \times q_0] [q_0 \neq q_1] \quad \text{--- (9)}$$

$$[q_0 \times q_0] \rightarrow 0 [q_0 \times q_0] [q_0 \times q_0] \mid 0 [q_0 \times q_1] [q_1 \times q_0] \quad \text{--- (10)}$$

$$[q_0 \times q_1] \rightarrow 0 [q_0 \times q_1] [q_1 \times q_1] \mid 0 [q_0 \times q_0] [q_0 \times q_1] \quad \text{--- (11)}$$

$$\rightarrow 0 [q_0 \times q_1] [q_1 \times q_0] \quad \text{--- (12)}$$

$$\rightarrow 0 [q_0 \times q_1] [q_1 \times q_1] \quad \text{--- (13)}$$

Start symbol S .

Tracing Generating:

00011

$$S \rightarrow [q_0 \neq q_1] \quad \text{--- (1)}$$

$$\rightarrow 0 [q_0 \times q_1] [q_1 \neq q_1] \quad \text{--- (9)}$$

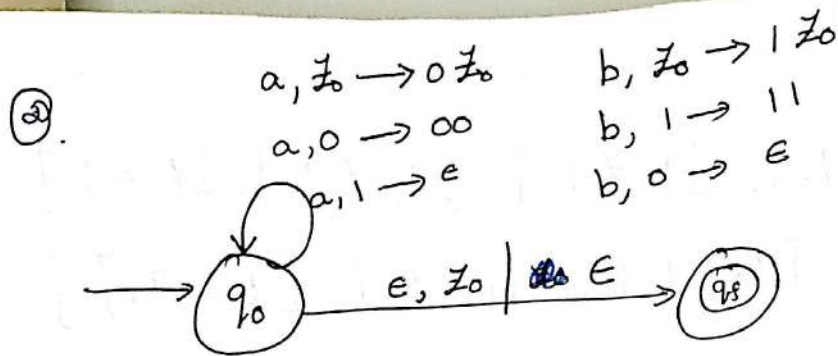
$$\rightarrow 0 0 [q_0 \times q_1] [q_1 \times q_1] [q_1 \neq q_1] \quad \text{--- (13)}$$

$$\rightarrow 0 0 0 [q_0 \times q_1] [q_1 \times q_1] [q_1 \neq q_1] \quad \text{--- (13)}$$

$$\rightarrow 0 0 0 1 [q_1 \times q_1] [q_1 \neq q_1] \quad \text{--- (3)}$$

$$\rightarrow 0 0 0 1 1 [q_1 \neq q_1] \quad \text{--- (4)}$$

$$\rightarrow 0 0 0 1 1 \epsilon \quad \text{--- (6)}$$



Using Rule ①:

$$S \rightarrow [q_0 \neq q_0] \mid [q_0 \neq q_f] \quad - ②$$

Using Rule ②:

$$\delta(q_0, a, 1) = (q_0, \epsilon) \Rightarrow [q_0 \mid q_0] \rightarrow a \quad - ③$$

$$\delta(q_0, b, 0) = (q_0, \epsilon) \Rightarrow [q_0 \mid q_0] \rightarrow b \quad - ④$$

$$\delta(q_0, \epsilon, z_0) = (q_f, \epsilon) \Rightarrow [q_0 \neq q_f] \rightarrow \epsilon \quad - ⑤$$

Using Rule ③:

①. $\delta(q_0, a, z_0) = (q_0, 0z_0)$

$$[q_0 \neq q_0] \rightarrow a [q_0 \mid q_0] [q_0 \neq q_0] \mid \quad - ⑥$$

$$a [q_0 \mid q_f] [q_f \neq q_0] \quad - ⑦$$

$$[q_0 \neq q_f] \rightarrow a [q_0 \mid q_0] [q_0 \neq q_f] \mid \quad - ⑧$$

$$a [q_0 \mid q_f] [q_f \neq q_f] \quad - ⑨$$

②. $\delta(q_0, b, z_0) = (q_0, 1z_0)$

$$[q_0 \neq q_0] \rightarrow b [q_0 \mid q_0] [q_0 \neq q_0] \mid \quad - ⑩$$

$$b [q_0 \mid q_f] [q_f \neq q_0] \quad - ⑪$$

$$[q_0 \neq q_f] \rightarrow b [q_0 \mid q_0] [q_0 \neq q_f] \mid \quad - ⑫$$

$$b [q_0 \mid q_f] [q_f \neq q_f] \quad - ⑬$$

$$(iii) \delta(q_0, a, 0) = (q_0, 00)$$

$$[q_0 \circ q_0] \rightarrow a [q_0 \circ q_0] [q_0 \circ q_0] \quad - (14)$$

$$a [q_0 \circ q_f] [q_f \circ q_f] \quad - (15)$$

$$[q_0 \circ q_f] \rightarrow a [q_0 \circ q_0] [q_0 \circ q_0] \quad - (16)$$

$$a [q_0 \circ q_f] [q_f \circ q_f] \quad - (17)$$

$$(iv) \delta(q_0, b, 1) = (q_0, 11)$$

$$[q_0 \mid q_0] \rightarrow b [q_0 \mid q_0] [q_0 \mid q_0] \quad - (18)$$

$$b [q_0 \mid q_f] [q_f \mid q_f] \quad - (19)$$

$$[q_0 \mid q_f] \rightarrow b [q_0 \mid q_0] [q_0 \mid q_f] \quad - (20)$$

$$b [q_0 \mid q_f] [q_f \mid q_f] \quad - (21)$$

Generating:

abab

$$S \rightarrow [q_0 \neq q_f] \quad (2)$$

$$\rightarrow a [q_0 \circ q_0] [q_0 \neq q_f] \quad (3)$$

$$\rightarrow ab [q_0 \neq q_f] \quad (4)$$

$$\rightarrow ab a [q_0 \circ q_0] [q_0 \neq q_f] \quad (5)$$

$$\rightarrow abab [q_0 \neq q_f] \quad (6)$$

$$\rightarrow abab \epsilon \quad (7)$$

$$(3). M = (\{q_0, q_1\}, \{a, b\}, \{A, B, z\}, \delta, q_0, z)$$

where δ is defined as,

$$\delta(q_0, a, z) = (q_0, Az)$$

$$\delta(q_0, b, z) = (q_0, Bz)$$

$$\delta(q_0, a, A) = \{(q_0, AA), (q_1, \epsilon)\}$$

$$\delta(q_0, b, B) = \{(q_0, BB), (q_1, \epsilon)\}$$

$$\delta(q_0, a, B) = \{(q_0, AB)\}$$

$$\delta(q_0, b, A) = \{(q_0, BA)\}$$

$$\delta(q_1, a, A) = (q_1, \epsilon)$$

$$\delta(q_1, b, B) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z) = (q_1, \epsilon)$$

Solution:

$$G = (V, T, P, S)$$

$$V = \{S, [q_0 z q_0], [q_0 A q_0], [q_0 B q_0], [q_1 z q_0], [q_1 B q_0], [q_1 A q_0], [q_0 z q_1], [q_0 A q_1], [q_0 B q_1], [q_1 A q_1], [q_1 B q_1], [q_1 z q_1]\}$$

$$T = \{a, b\}$$

$$P =$$

$$\{ S \rightarrow [q_0 z q_0] \mid [q_0 z q_1] \}$$

$$\left. \begin{array}{l} [q_0 z q_0] \rightarrow a [q_0 A q_0] [q_0 z q_0] \\ \quad \rightarrow a [q_0 A q_1] [q_1 z q_0] \\ [q_0 z q_1] \rightarrow a [q_0 A q_0] [q_0 z q_1] \\ \quad \rightarrow a [q_0 A q_1] [q_1 z q_1] \end{array} \right\} Az$$

$$[q_0 A q_0] \rightarrow a [q_0 A q_0] [q_0 A q_0]$$

$$\rightarrow a [q_0 A q_1] [q_1 A q_0]$$

$$[q_0 A q_1] \rightarrow a [q_0 A q_0] [q_0 A q_1]$$

$$\rightarrow a [q_0 A q_1] [q_1 A q_1]$$

AA

$$[q_0 Z q_0] \rightarrow b [q_0 B q_0] [q_0 Z q_0]$$

$$\rightarrow b [q_0 B q_1] [q_1 Z q_0]$$

$$[q_0 Z q_1] \rightarrow b [q_0 B q_0] [q_0 Z q_1]$$

$$\rightarrow b [q_0 B q_1] [q_1 Z q_1]$$

BZ

$$[q_0 B q_0] \rightarrow b [q_0 B q_0] [q_0 B q_0]$$

$$\rightarrow b [q_0 B q_1] [q_1 B q_0]$$

$$[q_0 B q_1] \rightarrow b [q_0 B q_0] [q_0 B q_1]$$

$$\rightarrow b [q_0 B q_1] [q_1 B q_1]$$

BB

$$[q_0 B q_0] \rightarrow a [q_0 A q_0] [q_0 B q_0]$$

$$\rightarrow a [q_0 A q_1] [q_1 B q_0]$$

$$[q_0 B q_1] \rightarrow a [q_0 A q_0] [q_0 B q_1]$$

$$\rightarrow a [q_0 A q_1] [q_1 B q_1]$$

AB

$$[q_0 A q_0] \rightarrow b [q_0 B q_0] [q_0 A q_0]$$

$$\rightarrow b [q_0 B q_1] [q_1 A q_0]$$

$$[q_0 A q_1] \rightarrow b [q_0 B q_0] [q_0 A q_1]$$

$$\rightarrow b [q_0 B q_1] [q_1 A q_1]$$

BA

$$[q_0 A q_1] \rightarrow a$$

$$[q_0 B q_1] \rightarrow b$$

$$[q_1 A q_1] \rightarrow a \quad [q_1 Z q_1] \rightarrow G$$

$$[q_1 B q_1] \rightarrow b$$

Decision Properties of the Regular Language:

- * Membership Problem
- * Emptiness of the Language
- * Finiteness of the Language

Ex for membership problem:

Given a regular expression r and a string x , does x belong to the language corresponding to r ?

Ex for Emptiness of the language:

Given a finite automaton M , is there a string that it accepts? (Alternatively, given an FA M , is $L(M) = \emptyset$?)

Ex for Finiteness of the language:

Given an FA M , is $L(M)$ finite?

①. Given 2 FAs M_1 and M_2 , is $L(M_1)$ a subset of $L(M_2)$?

Find $L(M_1) \subseteq L(M_2)$

$L(M_1) \subseteq L(M_2)$ if & only if $L(M_1) - L(M_2) = \emptyset$.

②. Given 2 FAs M_1 and M_2 , are there any strings that are accepted by neither?

Find M_1' and M_2' accepting $L(M_1)'$ & $L(M_2)'$

and then find $L(M_1') \cap L(M_2')$

- ③. Given an FA M accepting a language L , and given 2 strings x and y are x and y distinguishable with respect to L ?

x & y are distinguishable with respect to L if and only if $\delta_1^*(q_1, x) \neq \delta_1^*(q_1, y)$

For each statement below, decide whether it is true or false. If it is true, prove it. If not, give a counterexample. All parts refer to languages over the alphabet $\{0, 1\}$.

- ✓ (a). If $L_1 \subseteq L_2$ and L_1 is not regular, then L_2 is not regular. $L_1 - L_2 = \emptyset$

False. Σ^* has a nonregular subset.

- (b). If $L_1 \subseteq L_2$ and L_2 is not regular, then L_1 is not regular.

False. Non-regular language have finite subsets and finite languages are regular.

- ✓ (c). If L_1 and L_2 are non-regular, then $L_1 \cup L_2$ is non-regular. False. The union of any language and its complement is Σ^* , which is regular.

- ✓ (d). If L_1 and L_2 are nonregular, then $L_1 \cap L_2$ is non-regular. False. The intersection of a nonregular language and its complement is empty and the empty language is regular.

- ✓ (e). If L is non-regular, then L' is nonregular. True. The complement of a regular lang. is regular.

(P). If L_1 is regular and L_2 is nonregular, then $L_1 \cup L_2$ is nonregular.

False. L_2 could be a subset of L_1 .

(Q). If L_1 is regular, L_2 is nonregular, and $L_1 \cap L_2$ is regular, then $L_1 \cup L_2$ is nonregular.

True. $L_1 \cap L_2$ is reg. $L_1 - (L_1 \cap L_2) = L_1 - L_2$

$L_2 = (L_1 \cup L_2) - (L_1 - L_2) \therefore L_1 \cup L_2$ reg. L_2 is also reg.

(R). If L_1 is regular, L_2 is nonregular, and $L_1 \cap L_2$ is non-regular, then $L_1 \cup L_2$ is non-regular.

False. L_1 could be Σ^*

(S). If $L_1, L_2, L_3, \dots$ are all regular, then $\bigcup_{n=1}^{\infty} L_n$ is regular.

False.

$\{0^n 1^n \mid n \geq 0\} = \{1\} \cup \{01\} \cup \{0011\} \cup \dots$

(T). If $L_1, L_2, L_3, \dots$ are all non-regular and $L_i \subseteq L_{i+1}$ for each i , then $\bigcup_{n=1}^{\infty} L_n$ is nonregular.

False.

For each $k \geq 1$

let $S_k = \{0^{2^k i}, 1^{2^k i} \mid i \geq 0\}$

$S_1 = \{1, 0^2, 1^2, 0^4, 1^4, 0^6, 1^6, \dots\}$

$S_2 = \{1, 0^4, 1^4, 0^8, 1^8, \dots\}$

$S_{k+1} \subseteq S_k$ for each k .

The set $\bigcap_{k=0}^{\infty} S_k$ is $\{1\}$ which is regular but it is easy to show that for every k S_k is nonregular.

Now let $L_k = S_k'$. It follows that L_k is non-regular and $L_k \subseteq L_{k+1}$ but

$$\bigcup_{k=1}^{\infty} L_k = \left(\bigcap_{k=1}^{\infty} S_k \right)' = \Sigma^* - \{ \wedge \} \text{ and}$$

this set is regular.

CLOSURE PROPERTIES OF REGULAR LANGUAGES

- (*) The union of two regular languages is regular.
- (*) The intersection of two regular languages is regular.
- (*) The complement of a regular language is regular.
- (*) The difference of two regular languages is regular.
- (*) The reversal of a regular language is regular.
- (*) The closure (star) of a regular language is regular.
- (*) The concatenation of a regular language is regular.
- (*) A homomorphism (substitution of strings for symbols) of a regular language is regular.
- (*) The inverse homomorphism of a regular language is regular.
- (*) If L and M are regular languages, then so is LM .

Proof:

L and M are regular languages $\therefore$ they have reg. expressions

$$\Rightarrow L = L(R) \quad ; \quad M = L(S).$$

Then $LM = L(R+S)$ by the def. of the $+$ operator for regular expression.

- (*) If L is a regular language over alphabet Σ , then $\bar{L} = \Sigma^* - L$ is also a regular language.

Proof:

Let $L = L(A)$ for some DFA $A = (Q, \Sigma, \delta, q_0, F)$.

$\bar{L} = L(B)$ for $B = (Q, \Sigma, \delta, q_0, Q-F)$.

B is exactly like A , but the accepting states of A

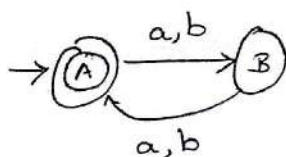
have become non-accepting states of B & vice versa.

$\Rightarrow w$ is in $L(B)$ if & only if $\delta(q_0, w)$ is in $Q-F$,

which occurs if & only if w is not in $L(A)$.

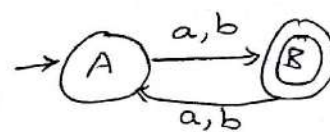
Ex:

①. $L_1 = \{ \text{strings of even length} \}$

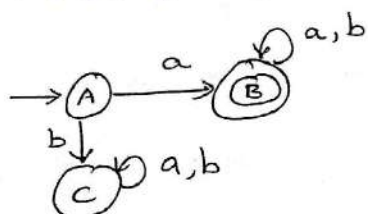


$$L_2 = \overline{L_1}$$

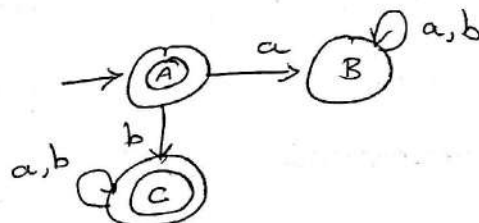
$= \{ \text{strings of odd length} \}$



②. $L_3 = \{ \text{strings starting with 'a'} \}$



$$L_4 = \overline{L_3}$$



③. If L and M are regular languages, then so is $L \cap M$.

Proof:

L and M are regular languages (given).

$$A_L = (Q_L, \Sigma, \delta_L, q_L, F_L)$$

$$A_M = (Q_M, \Sigma, \delta_M, q_M, F_M)$$

$\Sigma \Rightarrow$ assume it to be same

if it is diff then take union.

It works for both NFA and DFA.

$$A_{L \cap M} = (Q_L \times Q_M, \Sigma, \delta, (q_L, q_M), F_L \times F_M)$$

$$\text{where } \delta((p, q), a) = (\delta_L(p, a), \delta_M(q, a))$$

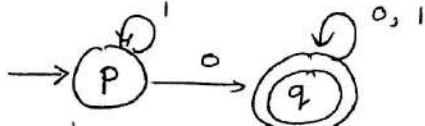
$$\therefore L(A_{L \cap M}) = L(A_L) \cap L(A_M)$$

$$\hat{\delta}((q_L, q_M), w) = (\hat{\delta}(q_L, w), \hat{\delta}(q_M, w))$$

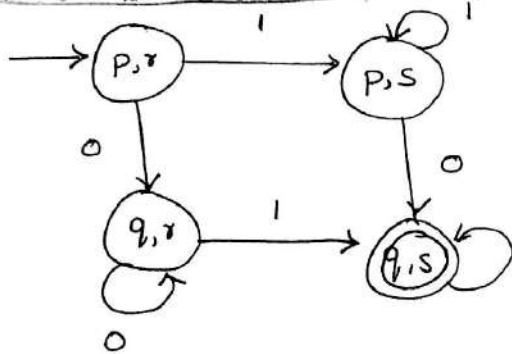
w is accepted by A if & only if both A_L and A_M accept w .

Thus A accepts the intersection of L and M .

Ex: $L = 1^* 0 (0+1)^*$ $L^R = (0+1)^* 0 1^*$



Given a language L that is $L(A)$ for some finite automaton, perhaps with non-determinism and ϵ -transitions, we may construct an automaton for L^R by:



- ①. Reverse all arcs in the transition diagram for A .
- ②. Make the start state of A be the only accepting state for the new automaton.
- ③. Create a new start state p_0 with transitions on ϵ to all the accepting states of A .

*4. If L and M are regular languages, then so is $L-M$.

Proof:

$$L - M = L \cap \bar{M}$$

$\bar{M}$ is regular and $L \cap \bar{M}$ is also regular

so $L-M$ is also regular.

*5. If L is a regular language, so is L^R .

Proof:

$$\begin{array}{ccc} L \Rightarrow & \epsilon & L^R = \epsilon \\ & \phi & \phi \\ & a & a \end{array}$$

$\epsilon \Rightarrow$ Regular Exp.

$$\therefore L = L^R$$

$$L = L_1 + L_2 \quad \text{then} \quad L^R = L_1^R + L_2^R$$

$$L = L_1 \cdot L_2 \quad \text{then} \quad L^R = L_2^R \cdot L_1^R$$

$$L(E_1) = \{01, 111\} \quad L(E_2) = \{00, 10\}$$

$$L(E_1 E_2) = \{0100, 0110, 11100, 11110\}$$

$$(L(E_1 E_2))^R = \{0010, 0110, 00111, 01111\}$$

$$L(E_2)^R \cdot L(E_1)^R = \{00, 01\} \cdot \{10, 111\} = \{0010, 00111, 0110, 01111\}$$

$$L = L_1^* \quad \text{then} \quad L^R = (L_1^R)^*$$

Using Kleene's Theorem

Id. pt of

re

7

1

$$w \in L(E)$$

$$w_i = w_1 w_2 \dots w_n$$

$$w^R = w_n^R w_{n-1}^R \dots w_1^R$$

Each w_i^R is in $L(E^R)$ so w^R is in $(L(E^R))^*$.

Any string in $L(E, R)^*$ is of the form $w_1 w_2 \dots w_n$, where each w_i is the reversal of a string in $L(E)$.

Reversal of the string $w_n^R w_{n-1}^R \dots w_1^R$ is $\therefore$ a string in $L(E, *)$ which is $L(E)$.

Thus string is in $L(E)$ if & only if its reversal is in $L(E, R)^*$.

$$\text{Ex: } R: L = (0+1)0^*$$

$$\bar{L} = 0^*(0+1)$$

(*) If L is a regular language over alphabet Σ , and h is a homomorphism on Σ , then $h(L)$ is also regular.

Proof:

Let Σ and Γ are set of alphabets.

$$\begin{array}{ccc} h: \Sigma & \rightarrow & \Gamma^* \\ \downarrow & & \downarrow \\ \text{single alph} & & \text{string} \end{array}$$

$$\text{If } w = a_1 a_2 a_3 \dots a_n$$

$$h(w) = h(a_1) h(a_2) \dots h(a_n)$$

If L is made of alphabets from Σ , then

$h(L) = \{h(w) \mid w \in L\}$ is called homomorphic image.

Ex:

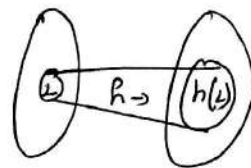
①. $\Sigma = \{0, 1\}$

$\Gamma = \{0, 1, 2\}$

What is $h(010)$?

$$h(010) = h(0)h(1)h(0) \\ = 011201$$

$h(0) = 01 \rightarrow$ string
 $h(1) = 112 \rightarrow$



②. $\Sigma = \{0, 1\}$

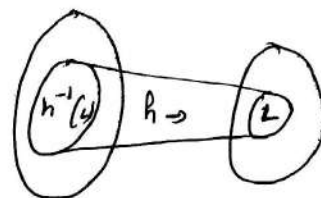
$\Gamma = \{1, 2, 3\}$

What is $h(010)$?

$$h(010) = 31221323122$$

$h(0) = 3122$

$h(1) = 132$



*9. If h is a homomorphism from alphabet Σ to alphabet Γ and L is a regular language over Γ , then $h^{-1}(L)$ is also a regular language.

Proof: [If L is regular & h is homomorphism, then homomorphic image $h(L)$ is regular]

Let R be reg. exp & $L(R)$ is reg. lang.

$h(R)$ can be found using $h(a)$ for each a in Σ .

By def. $h(R)$ is a reg. exp. & so $h(L)$ is reg. lang.

$\therefore$ Reg. lang. is closed under homomorphism.

Ex: If $L = \{00, 010\}$ what is homomorphic

① image of L ?

$$L(00, 010) = L(h(00), h(010)) \\ = L(h(0)h(0), h(0)h(1)h(0)) \\ = (0101, 011201)$$

② $h(0) = 3122$ $h(1) = 132$

$$(0+1)^* (00)^* = (h(0) + h(1))^* (h(0)h(0))^* \\ = (3122 + 132)^* (31223122)^*$$

Ex. Suppose h is the homomorphism from the alphabet $\{0, 1, 2\}$ to the alphabet $\{a, b\}$ defined by:
 $h(0) = a$; $h(1) = ab$; $h(2) = ba$.

(a). What is $h(0120)$? $aabbaa$

(b). What is $h(21120)$? $baababbaa$

(c). If L is the lang. $L(01^*2)$ what is $h(L)$? $a(ab)^*ba$.

(d). $L(0+12)$ " $(a + abba)$

(e). Suppose L is the language $\{ababa\}$, that is, the language consisting only the one string $ababa$. What is $h^{-1}(L)$?

$$\begin{array}{ccc} ababa & ababa & ababa \\ \hline 1 & 1 & 0 \\ \hline 1 & 0 & 2 \end{array}$$

$$h^{-1}(L) = \{110, 022, 102\}$$

Chomsky Hierarchy:

(1). Type 3 :
(Regular Grammar)

$A \rightarrow \alpha B | \beta$ Light Linear Grammar
 $A, B \in V$ $\Leftarrow$ Right Linear Grammar
 $\alpha, \beta \in T^*$

$A \rightarrow B\alpha | \beta$
 $A, B \in V$
 $\alpha, \beta \in T^*$

(2). Type 2 :
(CFG)

$A \rightarrow \alpha$, $A \in V$
 $\alpha \in (V \cup T)^*$

(3). Type 1 :
(CSG)

$\alpha \rightarrow \beta$, $|\beta| \geq |\alpha|$